ARTIFICIAL INTELLIGENCE–INTEGRATED RESEARCH ARCHIVAL SYSTEM FOR SOUTHERN LUZON TECHNOLOGICAL COLLEGE FOUNDATION, INC.

A Project Study Presented to the Faculty of
Master in Information Technology
Aemilianum College Inc.
Rizal St., Piot, West District, Sorsogon City
Sorsogon, Philippines 4700

In Partial Fulfillment
of the Requirements for the Degree
MASTER IN INFORMATION TECHNOLOGY

MARK IAN CABRIA VIBAR

May 2026

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

RECOMMENDATION FOR ORAL DEFENSE

In partial fulfillment of the requirements for the degree of MASTER IN INFORMATION TECHNOLOGY, this research project entitled, "**Artificial Intelligence-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc.,**" by **Mark Ian Cabria Vibar**, is hereby submitted to the thesis committee for oral examination.

MARCO L. ESPINOSA, MIT
Adviser

THESIS COMMITTEE

In partial fulfillment of the requirements for the degree of MASTER IN INFORMATION TECHNOLOGY, this research project, entitled, "**Artificial Intelligence-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc.,**" prepared and submitted by **Mark Ian Cabria Vibar**, is hereby recommended for Oral Defense.

JOANNAH C. ARGOTE, MIT
Member

SHERRY MAE R. LLANDELAR, DIT
Member

MILAN E. BAUSA, MIT
Member

JOSEFINA R. SARMIENTO, MIT, PhD
Member

REV. FR. JOEREX P. ALONSO, CRS
Chairman

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

RESULT OF THE ORAL DEFENSE

Result of the Oral Examination for **MARK IAN CABRIA VIBAR**, candidate for the degree, MASTER IN INFORMATION TECHNOLOGY.

Project Title	:	**Artificial Intelligence-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc.,"**
Adviser	:	Marco L. Espinosa, MIT
Place	:	Aemilianum College Inc.
Date	:	January 25, 2026
Day	:	Sunday
Time	:	3:00 p.m.

APPROVED by the Committee on the Oral Examination with a grade of _____%.

JOANNAH C. ARGOTE, MIT
Member

SHERRY MAE R. LLANDELAR, DIT
Member

MILAN E. BAUSA, MIT
Member

JOSEFINA R. SARMIENTO, MIT, PhD
Member

REV. FR. JOEREX P. ALONSO, CRS
Chairman

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

APPROVAL SHEET

ACCEPTED and APPROVED in partial fulfillment of the requirements for the degree of MASTER IN INFORMATION TECHNOLOGY.

Comprehensive Examinations – PASSED.

REV. FR. REY GENARO M. MALABANAN, CRS
School Director

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

ACKNOWLEDGEMENT

This thesis, *Artificial Intelligence–Integrated Research Archival System For Southern Luzon Technological College Foundation, Inc*., represents not only the researcher's hard work but also a journey he did not walk alone. The researcher humbly expresses his sincere gratitude to all individuals who have offered guidance, support, and encouragement throughout the completion of this study. Their contributions have played an important role in making this research possible:

To the *Aemilianum College Inc*., the researcher is sincerely grateful for the opportunity to pursue his MIT degree and be part of the Aemilian community. This experience has greatly contributed to his academic growth and personal development, and he will always be thankful for this privilege.

The researcher extends his heartfelt appreciation to *Rev. Fr. Rey Genaro M. Malabanan, CRS*, Director of Aemilianum College Inc., for his leadership and inspiration. His commitment to excellence and service has guided the researcher in carrying out this study with dedication and purpose.

The researcher also gives his deepest gratitude to subject professor and Dean, *Dr. Josefina R. Sarmiento*, for her constant support, expert guidance, and patience. Her wisdom, encouragement, and genuine concern have been invaluable throughout this

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

research journey. The researcher is truly grateful for her mentorship and unwavering support. Her insightful feedback challenged the researcher to think critically and strengthened the overall quality of this study. Beyond academia, her kindness and approachability created an environment that fostered growth and confidence.

To his capstone project adviser, *Mr. Marco L. Espinosa, MIT*, for the continuous encouragement and support in completing this project. His guidance provided clarity during challenging phases and inspired the researcher to pursue innovative solutions. The researcher sincerely appreciates his patience, constructive feedback, and dedication, which greatly contributed to the success of this capstone project.

Sincere thanks are also extended to the respected thesis committee members – *Prof. Richard G. Rabulan, Prof. Marco L. Espinosa, Prof. Milan E. Bausa, Dr. Sherry Mae R. Llandelar, Dr. Josefina R. Sarmiento, Rev. Fr. Joerex P. Alonso, CRS and Rev. Fr. Mandee N. Batac, CRS* - for their time, constructive insights, and valuable suggestions which significantly improved the quality of this study. Their thoughtful critiques and expert guidance helped refine the research methodology and strengthened the overall findings. The researcher deeply appreciates their dedication and generosity in sharing their knowledge and expertise throughout this process.

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The researcher also wishes to thank the *IT experts* and *end users* who evaluated the system. Your honest feedback, comments, and recommendations were essential in enhancing the system and ensuring its effectiveness. Their active participation and insightful observations provided practical perspectives that greatly improved the system's usability. The researcher is truly grateful for their time, patience, and valuable contributions to this project.

To his parents, *Josephine Vibar and Christopher Vibar*, the researcher offers his deepest gratitude for their unwavering love, guidance, and sacrifices. Their constant support became the foundation upon which his dreams were built. Through every challenge and triumph, they remained his source of wisdom, strength, and inspiration. Their enduring belief in his abilities motivated him to persevere and pursue excellence. This accomplishment is a reflection of their steadfast dedication and the values they have instilled in him.

To his brothers - *Clint Michael Vibar, Kim Joseff Vibar, and John Kian Vibar* - the researcher conveys his heartfelt appreciation for their support, understanding, and companionship. Their encouragement, whether shown through kind words, shared laughter, or simple presence, provided him with strength throughout this journey. They have been constant reminders of resilience, unity, and the importance of family.

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Their belief in him fueled his determination to continue moving forward, even in difficult moments.

To his *relatives, friends, colleagues, students and love one*, the researcher extends his sincere gratitude for their encouragement, support, and positive influence throughout this endeavor. Their thoughtful words, motivation, and shared moments of learning contributed to his growth both personally and professionally. Whether offering guidance, cooperation, or simple gestures of kindness, each of them played a meaningful role in sustaining his commitment to this work. Their presence enriched his journey and strengthened his resolve to achieve this milestone.

Above all, the researcher dedicates this achievement to *God Almighty*, who blessed him with wisdom in times of uncertainty, strength in moments of weakness, and grace to continue moving forward. Without His guidance, this accomplishment would not have been possible.

M.I.C.V.

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

PROJECT STUDY ABSTRACT

This study focused on the development of an AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. (SLTCFI) to enhance research management processes. The system was designed using HTML, CSS, and JavaScript for the front-end, an SQL-based database for structured storage, and cloud storage with mobile accessibility for secure and flexible access. APIs for external academic databases, such as OpenAlex and Semantic Scholar, and NLP algorithms were integrated to provide intelligent research recommendations. The development followed the Agile Development Methodology, encompassing phases of Requirements Analysis, Design, Development, Testing and Debugging, Release, and Maintenance with User Feedback, ensuring active stakeholder involvement, iterative refinement, and alignment with institutional requirements.

The findings revealed that the system was successfully developed with key modules, including the AI-Based RRL Finder, Research Upload and Archiving, Role-Based Access Control, and Approval and Verification Modules. Security, usability, and workflow optimization features, such as role-based user authentication, encrypted storage, secure cloud integration, and real-time notifications, were effectively implemented to enhance system performance. The system was evaluated using the ISO/IEC 25010 software quality model, achieving an overall rating of 4.32, which

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

indicated that it exceeded expectations in terms of functionality, efficiency, reliability, usability, maintainability, and portability.

Based on these findings, the study concluded that the AI-Integrated Research Archival System successfully improved efficiency, accessibility, and organization of research management, provided secure and controlled access, and offered intelligent, AI-driven support for literature discovery. The system proved to be a highly effective and technically sound tool for all stakeholders.

It is recommended that the system continue to be utilized, regularly maintained, and enhanced, including expanding API integrations, conducting user training, performing routine security audits, monitoring performance, and optimizing mobile access. These steps will ensure the system remains scalable, secure, user-friendly, and capable of supporting efficient research management at SLTCFI.

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

TABLE OF CONTENTS

Page

Title Page i
Recommendation for Oral Defense. ii
Result of the Oral Defense iii
Approval Sheet iv
Acknowledgement. v
Project Study Abstract. viii
Table of Contents. x
List of Tables xii
List of Figures. xiv
List of Appendices xv

Chapter

I. **Project Background** 1
Introduction 1
General Objective. 4
Specific Objectives. 4
Scope and Delimitations 5
Significance of the Study. 7
Project Dictionary 8

II. **Review of Related Literature and Systems**. . . . 15
Related Literature 15
Related Systems. 29
Synthesis of the State-of-the-Art 47

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Gap Bridged by the Study 50
Conceptual Framework. 51

III. **Technical Background**. 55
Resources. 55
Stakeholders. 57
Constraints. 61
Project Development Methodology. 67

IV. **Artificial Intelligence–Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc.** . 71
Requirements Analysis 71
Design. 74
Development. 87
Testing and Debugging 104
Release. 120
Maintenance and Feedback 121

V. **Summary of Findings, Conclusions, and Recommendations** 122
Findings. 122
Conclusions 124
Recommendations. 125

Bibliography 128
Appendices. 134

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

LIST OF TABLES

Table No.	Title	Page
3.1.	Software Requirement.	55
3.2.	Hardware Requirement.	56
3.3.	Stakeholder.	59
3.4.	System Development Time Frame	63
3.5.	Estimated Cost of the System.	66
4.1.	The Evaluation Rubric.	106
4.2.	Mean Interpretation Scale for System Evaluation. .	108
4.3.	System Functionality.	109
4.4.	System Reliability.	110
4.5.	System Usability.	111
4.6.	System Efficiency.	112
4.7.	System Maintainability.	113
4.8.	System Portability.	114
4.9.	IT Experts' View on the Developed System. . .	116
4.10.	Users'/Stakeholders View on the Developed System. .	117

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

LIST OF FIGURES

Figure No.	Title	Page
2.1.	Conceptual Paradigm.	54
3.1.	Agile Software Development.	70
4.1.	Data Flow Diagram of the Existing System. . .	73
4.2.	Data Flow Diagram of the Developed System. . .	75
4.3.	Data Flow Diagram of Login Module. . . .	77
4.4.	The Data Flow Diagram (DFD) of the Creating Account Module.	79
4.5.	Data Flow Diagram of the Research Abstract Uploading Module.	81
4.6.	Data Flow Diagram of the Research Abstract Uploading Module.	83
4.7.	System Architecture of the Developed System. . .	85
4.8.	Home Page.	89
4.9.	Login Module.	90
4.10.	System Dashboard.	91
4.11.	The Administrator List of Campus Tab. . . .	92
4.12.	The Administrator List of Department Tab. . .	93
4.13.	The Administrator Account Validation Tab. . .	94

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

4.14. The Administrator System Settings Tab. . . . 95

4.15. The Deans' Dashboard. 96

4.16. The Deans Research Abstract Approval Tab. . . 97

4.17. The Deans User Monitoring Tab. 98

4.18. The Faculty Dashboard. 99

4.19. The Faculty Research Management Tab. . . . 100

4.20. The Faculty Student Account Management Tab. . . 101

4.21. The Student Dashboard. 102

4.22. The Student Abstract Uploading Tab. . . . 103
.

Aemilianum College Inc.
GRADUATE SCHOOL
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

LIST OF APPENDICES

Appendix	Title	Page
A	Communications.	135
B	Evaluation Tools	137
C	Source Codes	139
D	System Testing and Evaluation.	144
E	Curriculum Vitae	145

CHAPTER I

PROJECT BACKGROUND

This chapter presents the project context, purpose and description of the study, its general and specific objectives, scope and delimitation, the significance of the study and the project dictionary which major terms are defined conceptually, technically and operationally for better understanding.

Introduction

Academic knowledge management is being transformed by rapid advancements in digital technologies, particularly artificial intelligence and automated information systems. International research highlights how AI-driven archival platforms revolutionize the way institutions organize, preserve, and retrieve scholarly outputs. Tawalbeh (2024) emphasizes that AI-integrated repositories significantly enhance metadata accuracy, automate classification, and improve search precision in university archives. These findings underscore a worldwide shift in academic institutions toward intelligent, automated research management solutions capable of handling rapidly increasing volumes of data. Thus, adapting global advancements in intelligent archiving becomes indispensable for schools seeking to improve research efficiency and maintain a competitive, technology-driven learning environment.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Recent studies in the Philippines highlight the urgent need for fully digital and intelligent academic repositories. Dizon and Evangelista (2023) found that higher education institutions adopting AI-assisted document management systems experienced substantial improvements in record accuracy, transparency, and accessibility of research files. Likewise, De Guzman et al. (2024) emphasized that digital transformation in Philippine HEIs accelerates research workflows and reduces administrative delays, particularly when metadata generation and document retrieval processes are automated. By integrating AI features such as intelligent search and automated RRL retrieval, schools can significantly enhance their ability to organize, validate, and preserve scholarly work while aligning with national efficiency and transparency standards.

Local research also highlights the challenges faced by Bicol-based institutions in managing academic documents. Malto and Rosero (2023) noted that local colleges transitioning to centralized digital archives experienced major improvements in data accessibility and reduced document misplacement. The relevance of developing an AI-integrated archival system for SLTCFI. The challenges highlighted in these studies manual indexing, scattered archives, and lack of intelligent search mechanisms mirror the current issues experienced by the institution. By incorporating AI automation, the system would place SLTCFI at the forefront of research innovation within the region.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Southern Luzon Technological College Foundation, Inc. continues to face substantial challenges in managing its rapidly growing collection of research outputs. Research files remain distributed across different departments, resulting in difficulties in tracking submissions, verifying previous works, and retrieving studies efficiently. Students often struggle to locate related literature, faculty members experience delays in accessing previous research, and administrators face difficulties maintaining organized and updated archives. These inefficiencies hinder academic continuity, slow down research production, and limit institutional readiness for accreditation and quality assurance evaluations.

The Artificial Intelligence-Integrated Research Archival System for SLTCFI is designed to directly address these issues. By introducing AI-powered metadata extraction, intelligent search capabilities, and an automated RRL Finder connected to scholarly databases such as OpenAlex and Semantic Scholar, the system modernizes the school's research infrastructure. It provides a centralized, secure, and scalable platform for storing and retrieving academic outputs. Much like the evolution of document tracking systems in other institutions, this project serves as a strategic modernization effort, supported by an Agile methodology and evaluated through the ISO/IEC 25010 framework to ensure international standards of system quality, reliability, and usability.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

General Objective

This study focused on the development of the Artificial Intelligence–Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc.

Specific Objectives

The study sought to accomplish the following specific objectives:

1. Develop an AI-Integrated Research Archival System using:
    1.1. HTML, CSS, and JavaScript for front-end development.
    1.2. An SQL-based database for structured storage and efficient retrieval.
    1.3. Cloud storage and mobile accessibility for secure and flexible research management.
    1.4. APIs for integration with external academic databases, such as OpenAlex and Semantic Scholar.
    1.5. NLP algorithms to enhance AI-driven research recommendations.
2. Develop system modules that enhance research management, including:
    2.1. AI-Based RRL Finder Module
    2.2. Research Upload and Archiving Module
    2.3. Role-Based Access Control Module
    2.4. Approval and Verification Module
3. Implement security, usability, and workflow optimization features, including:

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

3.1. User authentication with role-based access

3.2. Encrypted storage and secure cloud integration

3.3. Real-time notifications for document updates and approvals

4. Evaluate the developed system using the ISO/IEC 25010 software quality model in terms of:

4.1. Functionality

4.2. Efficiency

4.3. Reliability

4.4. Usability

4.5. Maintainability

4.6. Portability

Scope and Delimitations

The study focused on the development of an AI-Integrated Research Archival System designed to enhance research management at Southern Luzon Technological College Foundation, Inc. The system was developed using HTML, CSS, and JavaScript for the front-end, supported by an SQL-based database for structured storage and efficient data retrieval. It incorporated cloud storage and mobile accessibility for secure and flexible research management, APIs for integration with external academic databases such as OpenAlex and Semantic Scholar, and NLP algorithms to enhance AI-driven research recommendations. The system included modules to

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

support research management, namely the AI-Based RRL Finder Module, Research Upload and Archiving Module, Role-Based Access Control Module, and Approval and Verification Module. Key features also included user authentication with role-based access, encrypted storage with secure cloud integration, and real-time notifications for document updates and approvals. The system's quality was evaluated using the ISO/IEC 25010 software quality model, focusing on functionality, efficiency, reliability, usability, maintainability, and portability, with assessments conducted by ten end users and ten IT experts.

This study was limited to the development and evaluation of the AI-Integrated Research Archival System and did not cover research management processes outside the digital environment of Southern Luzon Technological College Foundation, Inc. Only the specified system modules and features were implemented, and evaluation was confined to selected evaluators, namely administrative staff, IT experts, and private institution representatives. The system was designed primarily for structured academic research documentation and did not extend to non-academic or informal research archives. Furthermore, while cloud storage and mobile access were implemented, the study did not explore other third-party cloud platforms or alternative AI technologies beyond the NLP algorithms incorporated. The evaluation focused on software quality attributes defined by ISO/IEC 25010, without measuring

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

long-term adoption, user behavior patterns, or institutional impact beyond the assessment period.

Significance of the Study

The study could be important to the following in particular:

Researchers and Students. They will benefit from an organized, AI-powered system that streamlines research retrieval, recommendation, and archival, making literature review and research documentation faster and more efficient.

Faculty Members. The system will assist faculty in managing, reviewing, and approving student and institutional research outputs with ease, improving workflow and collaboration.

IT Staff and System Administrators. They will gain a more secure, maintainable, and structured system for managing research data, reducing manual workload and minimizing errors.

Academic Institutions. Southern Luzon Technological College Foundation, Inc., and similar institutions will benefit from a centralized digital repository that supports research integrity, accessibility, and collaboration.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Administrative Staff. The system provides tools for tracking, approving, and archiving research projects efficiently, saving time and ensuring organized records.

Private Institutions and Partner Organizations. These stakeholders can access validated research outputs for collaboration, benchmarking, or further study, ensuring a more reliable exchange of academic knowledge.

Future Researchers. The system serves as a reference model for developing similar AI-integrated research management platforms or enhancing existing systems.

Project Dictionary

The following terms were used in the study and are defined conceptually and operationally for better understanding:

Artificial Intelligence–Integrated Research Archival System. An AI-integrated research archival system is a digital platform that leverages artificial intelligence to manage, organize, and recommend research documents efficiently (Smith, 2021). Here, it refers to the system developed using AI algorithms, SQL database, cloud storage, and user-access modules to manage research papers at SLTCFI.

*Southern Luzon Technological College Foundation, Inc***.** Southern Luzon Technological College Foundation, Inc. (SLTCFI) is a higher education institution that provides technological and academic programs to foster student learning and

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

research (Dela Cruz, 2020). For the purpose of this research, it refers to the institution where the AI-integrated research archival system is implemented.

*Structured Storage***.** Structured storage is an organized method of storing data in a predefined format to ensure efficient access and management (Garcia, 2019). Here, it refers to the use of an SQL-based database to categorize and store research documents with fields like title, author, date, and keywords.

*Efficient Retrieval***.** Efficient retrieval is the ability to quickly locate and access relevant information from a large dataset (Johnson, 2018). In this research, it denotes the system's capacity to allow users to search and retrieve research papers using keywords, filters, and AI-driven recommendations.

Cloud Storage. Cloud storage is a service that allows data to be stored on remote servers accessible via the internet (Miller, 2020). For this project, it represents storing research files in a secure cloud platform to enable access across multiple devices.

*Mobile Accessibility***.** Mobile accessibility is the ability to access digital services and data using mobile devices (Lee, 2019). Here, it means that users can access research documents and system features using smartphones or tablets.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Research Management. Research management involves organizing, monitoring, and facilitating research activities to ensure systematic progress (Nguyen, 2021). In this context, it refers to the uploading, archiving, reviewing, and tracking of research outputs through the system.

External Academic Databases. External academic databases are digital repositories of scholarly publications accessible online (Patel, 2018). Here, it refers to platforms such as OpenAlex and Semantic Scholar, integrated into the system to provide references.

OpenAlex. OpenAlex is an open database of scholarly works, authors, and institutions used for academic research (Wang, 2021). For this research, it serves as a source of external references for the AI recommendation engine.

Semantic Scholar. Semantic Scholar is an AI-powered research database designed to improve scholarly search and discovery (Klein, 2020). Here, it is used to retrieve and suggest relevant academic articles for research purposes.

NLP Algorithms. Natural Language Processing (NLP) algorithms enable computers to understand and process human language (Chowdhury, 2020). For this system, NLP algorithms analyze research titles, abstracts, and keywords to provide AI-driven recommendations.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

System Modules. System modules are distinct functional components within a software application that perform specific tasks (Harrison, 2019). Here, it refers to the AI-Based RRL Finder, Research Upload and Archiving, Role-Based Access Control, and Approval and Verification modules.

AI-Based RRL Finder Module. An AI-based RRL Finder is a system module that uses AI to locate relevant related literature for research (Lopez, 2021). In this research, it denotes the feature that automatically suggests relevant references from internal and external databases.

Research Upload and Archiving Module. A research upload and archiving module allows users to submit and store documents in an organized repository (Singh, 2018). Here, it is the component that enables users to securely upload, categorize, and save research papers.

Role-Based Access Control Module. Role-based access control (RBAC) restricts system access based on user roles and permissions (Sandhu et al., 1996). For the purpose of this research, it ensures that students, faculty, and administrators have access only to authorized system features.

Approval and Verification Module. An approval and verification module ensures submitted content meets standards before acceptance (Turner, 2017). Here,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

it refers to the module that allows authorized personnel to review, approve, and verify research documents prior to archiving.

Workflow Optimization. Workflow optimization improves efficiency by streamlining tasks and reducing redundant processes (Davenport, 2018). In this context, it refers to features like automated notifications, task tracking, and approval workflows that enhance research management.

User Authentication. User authentication is the process of verifying the identity of a user before granting system access (Stallings, 2019). Here, it denotes the login mechanism requiring valid credentials to access system features according to user roles.

Encrypted Storage. Encrypted storage protects data by converting it into an unreadable format for unauthorized users (Menezes et al., 2018). For this research, it refers to the encryption of research documents before storing them in the database or cloud.

Secure Cloud Integration. Secure cloud integration ensures that cloud-based services are safely connected with internal systems (Kim & Solomon, 2018). Here, it refers to the safe connection between the system and cloud servers to protect data during storage and retrieval.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Real-Time Notifications. Real-time notifications inform users instantly about system events or updates (Brown, 2020). For the purpose of this research, it is the feature that alerts users immediately when documents are uploaded, approved, or modified.

Functionality. Functionality refers to the degree to which a software system performs its intended functions (ISO/IEC 25010, 2011). Here, it is evaluated based on the system's ability to perform tasks such as document uploading, retrieval, and AI recommendations.

Efficiency. Efficiency is the capability of a system to perform tasks with optimal resource usage (ISO/IEC 25010, 2011). For this research, it is measured through response time, processing speed, and resource utilization.

Reliability. Reliability is the ability of a system to operate under specified conditions without failure (ISO/IEC 25010, 2011). Here, it is assessed through system uptime, consistency, and error rates during research management operations.

Usability. Usability is the ease with which users can learn and effectively use a system (ISO/IEC 25010, 2011). For the purpose of this research, it is determined through user feedback on navigation, interface design, and task completion ease.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Maintainability. Maintainability is the degree to which a system can be modified to correct faults or improve performance (ISO/IEC 25010, 2011). Here, it refers to the ease of updating modules, fixing bugs, and adding features.

Portability. Portability is the ease with which software can be transferred to different environments (ISO/IEC 25010, 2011). For this research, it denotes the system's ability to function across multiple devices and platforms, including desktops, tablets, and mobile phones.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This section discusses the reviewed related literature, including its relevance and significance to the study. It also highlights the similarities and differences among the reviewed studies. Furthermore, the synthesis of the state of the art, the research gap addressed by the study, and the conceptual framework are presented in this section.

Related Literature

Presented in this section are the foreign and local literatures that have significance and relevance to the present study.

Foreign

Artificial Intelligence (AI) refers to the branch of computer science that enables machines to perform tasks that typically require human intelligence, such as learning, reasoning, problem-solving, and decision-making. AI systems are designed to analyze data, recognize patterns, and make predictions or recommendations, making them valuable tools in various fields, including education, healthcare, and research management (Russell & Norvig, 2021).

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The reviewed articles are significant to the present study as they establish the fundamental role of Artificial Intelligence in performing complex tasks such as data analysis, pattern recognition, and decision-making, which are traditionally dependent on human intelligence. Their emphasis on balancing AI automation with human oversight is particularly relevant, as it underscores the need for accuracy, accountability, and ethical implementation in AI-based systems. This directly informs the design of the AI-Integrated Research Archival System for SLTCFI, which adopts a hybrid approach by integrating AI-driven features such as intelligent search, metadata extraction, and automated recommendations while maintaining faculty and administrator validation. In this context, the literature provides a critical foundation for ensuring that the system not only enhances efficiency and accessibility of research documents but also preserves reliability, data integrity, and responsible use of AI in academic research management.

The Artificial Intelligence can significantly enhance the functionality of digital archives in university libraries by automating tasks such as metadata generation, document classification, and retrieval processes. The article demonstrated that AI reduces human error, improves search accuracy, and increases overall accessibility to academic materials (Tawalbeh, 2024).

This literature is relevant and significant to the present study as it supports the use of Artificial Intelligence to streamline research management and improve the

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

efficiency of locating and accessing academic information. Its findings validate the role of automation in enhancing academic repositories, which directly aligns with the objectives of the AI-Integrated Research Archival System for SLTCFI. While both the reviewed study and the present research share the common goal of improving academic repositories through automation, the SLTCFI system extends the significance of these concepts by incorporating real-time notifications, mobile accessibility, and cloud-based storage through Firebase. These enhancements allow for continuous access, real-time monitoring, and more effective management of research outputs, thereby strengthening system usability, scalability, and overall institutional research support.

The responsible AI practices in libraries and archives require clear policies on algorithmic transparency, data privacy, and ethical handling to prevent misuse of academic content. The article highlighted that the rapid adoption of AI tools necessitates greater accountability from institutions to ensure accurate, fair, and ethical representation of information (Mannheimer, 2024).

The reviewed article is highly relevant and significant to the present study as it emphasizes the importance of implementing Artificial Intelligence in a controlled, ethical, and responsible manner within library and archival systems. Although Mannheimer's study is situated in a broader global context, its focus on ethical governance, transparency, and accountability provides a critical conceptual framework

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

for the AI-Integrated Research Archival System for SLTCFI. These principles directly inform the system's design and management, particularly in handling sensitive student research outputs. In this regard, the literature strengthens the present study by guiding the responsible use of AI, ensuring data privacy, trustworthiness, and institutional accountability in the development of an AI-assisted research repository.

The users of digital archives are often unaware of the extent to which AI is involved in data processing and retrieval, which can lead to misconceptions about the accuracy and reliability of the system. Her article highlights the importance of educating users and raising awareness to ensure the effective adoption of AI-powered tools (Jaillant, 2024).

This insight is relevant and significant to the present study as it highlights the importance of user orientation and digital literacy in the effective adoption of Artificial Intelligence within archival and research systems. Jaillant's study emphasizes that users must understand how AI functions in order to fully benefit from its capabilities, particularly in enhancing research discovery and efficiency. This directly supports the objectives of the AI-Integrated Research Archival System for SLTCFI, which recognizes that technological effectiveness depends not only on system features but also on user competence and engagement. While both Jaillant's study and the present system aim to improve user experience through AI, the SLTCFI system extends this insight by integrating training modules and a user-friendly interface tailored to students and

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

faculty, thereby promoting informed usage, usability, and sustained adoption within the academic community.

Spina (2023) highlighted that integrating Artificial Intelligence tools, such as ChatGPT, into archival and historical research workflows can greatly improve tasks like indexing, summarization, and document analysis. However, she cautioned that excessive reliance on AI may compromise the contextual understanding of archival content.

The abovementioned article is relevant and significant to the present study as it supports the use of Artificial Intelligence in enhancing academic efficiency through automated literature discovery. This is particularly applicable to the AI-Integrated Research Archival System for SLTCFI, specifically its AI-driven RRL Finder, which facilitates faster and more efficient identification of related studies. The article's emphasis on balancing automation with human involvement reinforces the system's design principle of requiring human verification to ensure contextual accuracy and relevance. While both studies demonstrate how AI can support academic work, the SLTCFI system is significant in that it deliberately limits automation to assist rather than replace human decision-making, thereby preserving scholarly oversight, reliability, and the academic integrity of the research process.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Jaillant (2025) emphasized that in the AI era, data are no longer static records but dynamic resources that require continuous updating and re-indexing. Her topic in *AI & Society* highlighted the need for modern archives to adopt adaptable architectures capable of accommodating evolving data and machine learning updates.

This article is relevant and significant to the present study as it emphasizes the importance of scalable database design and modular system architecture in managing growing digital archives. Its discussion on handling continuous data additions and system updates directly supports the development of the AI-Integrated Research Archival System for SLTCFI, which similarly requires a flexible and expandable infrastructure to accommodate increasing volumes of research outputs. Although Jaillant's work focuses on international archival infrastructures, its principles are highly applicable at the institutional level. The SLTCFI system operationalizes these concepts within a smaller academic repository, ensuring that future research materials can be seamlessly integrated while maintaining system efficiency, usability, and long-term sustainability.

Frontoni, et al emphasized that long-term data preservation in AI-powered archives depends on maintaining data integrity, authenticity, and verifiability through digital provenance tracking. Their article highlighted that archival preservation should not only store information but also protect it from alteration, corruption, or unauthorized access (Frontoni, Paolanti, Lauriault, Stiber, and Duranti, 2022).

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

This principle is directly relevant to the AI-Integrated Research Archival System for SLTCFI, which implements role-based access control, automated backups, and cloud synchronization to protect academic outputs. While both the reviewed approach and the SLTCFI system prioritize data security and authenticity, the SLTCFI system adapts these strategies to a localized academic context, ensuring that preservation measures are specifically tailored to the institution's research needs and operational requirements. This targeted application enhances both the reliability and practical usability of the archival system for students, faculty, and administrators.

Ocbian and Gamba highlighted that developing a records archiving and digital document repository system can significantly improve accessibility, accuracy, and overall efficiency in data management within academic institutions. Their article demonstrated that an electronic repository reduces redundancy and facilitates faster research retrieval (Ocbian and Gamba, 2014).

This is relevant to the AI-Integrated Research Archival System for SLTCFI, as it builds on similar objectives of improving document management and accessibility. While both systems aim to streamline research retrieval, the SLTCFI system distinguishes itself by incorporating AI-based functionalities such as intelligent keyword search, automated metadata tagging, and RRL recommendations thereby enhancing efficiency, accuracy, and the overall user experience beyond what traditional electronic repositories offer.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

According to ResearchGate, AI-driven open-access repositories have transformed knowledge sharing by improving discoverability, accessibility, and interconnectivity of research data. The article emphasized that AI enables smarter linking between related studies, reducing the time and effort required for researchers to locate relevant resources (ResearchGate, 2025).

This literature is relevant to the AI-Integrated Research Archival System for SLTCFI, as it supports the implementation of an AI-powered RRL Finder that automatically retrieves related studies from the institution's academic repository, enhancing research efficiency and facilitating easier access to relevant academic resources.

The author emphasized that advanced optical character recognition (OCR) techniques can preserve document layouts and extract structural metadata from complex archival materials, improving search accuracy and accessibility. (Beyene and Dancy, 2025).

This article by Beyene and Dancy is particularly relevant to the AI-Integrated Research Archival System for SLTCFI, as many older theses and research outputs exist only in scanned formats and require digital conversion. By incorporating OCR technology, the SLTCFI system can digitize these documents, ensuring long-term preservation and providing easier access for students, faculty, and administrators.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Local

Acebron (2024) emphasized that the effectiveness of digital archiving systems, such as those implemented in the Provincial Government of Laguna, relies heavily on user awareness, training, and management support. The article demonstrated that even highly advanced systems can fail if users lack the technical understanding or motivation to utilize digital tools effectively.

This literature is particularly relevant to the AI-Integrated Research Archival System for SLTCFI, as it highlights the importance of providing proper orientation and training for students, faculty, and administrators to ensure effective adoption and optimal use of the system..

Daluyon (2024) highlighted that designing and developing an electronic document archiving system for the University of Rizal System significantly improved the efficiency of document storage and retrieval. The topic also emphasized the importance of database normalization and data security protocols in preventing data duplication and ensuring data integrity.

This literature is significant to the AI-Integrated Research Archival System for SLTCFI, as it underscores the value of structured database design in maintaining accurate and reliable research records. The SLTCFI system builds on Daluyon's model by integrating AI-driven automatic classification and metadata generation, which

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

minimizes manual encoding errors, enhances search functionality, and improves the overall user experience, thereby offering a more efficient and intelligent approach to managing academic research outputs.

Cuevas (2022) emphasized that implementing an electronic document archiving system at Sto. Niño Catholic School streamlined record management and significantly reduced paper-based workloads. The topic highlighted the importance of transitioning from manual to digital systems to improve institutional productivity and accuracy in document tracking.

This literature is particularly relevant to the AI-Integrated Research Archival System for SLTCFI, as both share the goal of enhancing efficiency in information management. The SLTCFI system extends this approach by incorporating AI-powered retrieval and automated RRL suggestions, focusing specifically on academic research repositories rather than general administrative documents. This targeted application addresses the unique needs of students, faculty, and researchers, improving accessibility, usability, and the overall management of scholarly outputs.

Ocbian and Gamba's study is relevant to the present research as it establishes the value of developing a digital records archiving and document repository in an academic institution. Their article demonstrate that centralized digital systems

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

improve access to institutional data and research outputs, supporting efficiency and reliability in academic record management (Ocbian and Gamba, 2024).

Ocbian and Gamba's article is relevant to the present research because it highlights the benefits of establishing a digital records archiving and document repository within an academic institution. Their topic demonstrate that centralized digital systems enhance access to institutional data and research outputs, promoting efficiency, reliability, and systematic management of academic records. This directly informs the development of the AI-Integrated Research Archival System for SLTCFI by providing a foundational rationale for digitizing and centralizing academic resources, which supports streamlined retrieval, organized preservation, and improved accessibility for students, faculty, and administrators.

Caluza emphasized that the Electronic Document Archive Management System (EDAMS) developed for a university registrar's office effectively automated record management processes and reduced data redundancy. The article highlighted that implementing automation alongside controlled access levels enhances both data security and integrity (Caluza, 2025).

This is relevant to the AI-Integrated Research Archival System for SLTCFI, which builds on these principles by incorporating AI algorithms for intelligent data tagging and retrieval. By combining automation with AI-driven, context-aware functionalities,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

the SLTCFI system not only streamlines document management but also improves the research experience for students, faculty, and administrators, offering smarter and more efficient access to academic outputs.

IJMRA (2023) highlighted that incorporating SMS notification support in digital archiving systems enhances user interaction and accountability in document processing. The article found that real-time alerts improved user satisfaction and accelerated response times for document approvals (IJMRA, 2023).

This is particularly relevant to the AI-Integrated Research Archival System for SLTCFI, which incorporates similar real-time notifications for thesis approvals and review updates. While both systems aim to enhance communication and responsiveness, the SLTCFI system expands on this by providing a mobile-based interface, offering broader accessibility and convenience for students, faculty, and administrators within the academic environment.

Bicol University emphasized that implementing a digital archiving system for student records improved administrative efficiency and transparency in managing academic documents. The topic highlighted that centralized data storage and electronic tracking of record requests were key factors in the system's success (IJMRA, 2023).

The study from Bicol University is relevant to the present research as it demonstrates how implementing a digital archiving system can enhance

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

administrative efficiency and transparency in managing academic documents. Its emphasis on centralized data storage and electronic tracking of record requests underscores practical strategies for improving accessibility, organization, and accountability in record management. These insights inform the development of the AI-Integrated Research Archival System for SLTCFI by highlighting the benefits of a centralized, digitized repository, which can streamline the handling of research outputs and academic records while ensuring accurate tracking and efficient service for students, faculty, and administrators.

InnoCon Publishing emphasized that implementing electronic archiving systems in Philippine universities promotes long-term data sustainability and aligns with global digitization trends. The article highlighted that institutions adopting such systems experienced faster document retrieval and improved compliance with accreditation standards (InnoCon Publishing, 2024).

The study by InnoCon Publishing is relevant to the present research as it highlights the benefits of electronic archiving systems in promoting long-term data sustainability and aligning institutional practices with global digitization trends. The findings emphasize that such systems enable faster document retrieval and support compliance with accreditation standards, demonstrating both operational and regulatory advantages. This directly informs the AI-Integrated Research Archival System for SLTCFI, reinforcing the importance of a digital repository that not only

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

preserves academic outputs but also enhances accessibility, efficiency, and adherence to institutional and accreditation requirements.

De La Salle University emphasized that digitization and asset management of archival materials are essential for preserving institutional heritage and ensuring long-term data longevity. The literature highlighted the importance of efficient metadata management and digital asset preservation to prevent information loss over time (De La Salle University, 2023).

The study from De La Salle University is relevant to the present research as it underscores the importance of digitization and systematic asset management in preserving institutional heritage and ensuring the long-term longevity of academic data. The literature emphasizes efficient metadata management and proper digital asset preservation as critical strategies to prevent information loss over time. These insights inform the AI-Integrated Research Archival System for SLTCFI by highlighting the need for structured metadata, secure storage, and preservation practices that maintain the integrity, accessibility, and sustainability of research outputs for future academic use.

Philippine E-Journals highlighted that the emergence of institutional repositories in higher education has increased the visibility of research outputs and fostered collaboration among faculty and students. The topic noted that digital

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

repositories encourage research continuity, knowledge sharing, and academic innovation (Philippine E-Journals, 2024).

The article from Philippine E-Journals is relevant to the present research as it highlights how institutional repositories in higher education enhance the visibility of research outputs and foster collaboration among faculty and students. The literature emphasizes that digital repositories support research continuity, knowledge sharing, and academic innovation. These insights directly inform the AI-Integrated Research Archival System for SLTCFI by demonstrating the value of a centralized digital repository in promoting accessibility, collaboration, and the dissemination of scholarly work within the institution.

Related Systems

This section discusses existing systems that are relevant to the development of the School Research Archival System with Artificial Intelligence (AI) Integration. These systems, both foreign and local served as important references in understanding features and functionalities that contribute to effective research management, data accessibility, and system intelligence.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Foreign

Smith and Anderson examined the implementation of an AI-based institutional repository in a European university and found that intelligent search and automated metadata extraction significantly improved the discoverability of academic resources. Their study emphasized that AI-enhanced repositories reduce the time spent on manual indexing and enable researchers to retrieve more relevant literature efficiently, thereby strengthening research productivity and institutional knowledge management (Smith & Anderson, 2020).

Smith and Anderson's study was similar to the present research in that both focused on the use of Artificial Intelligence to improve institutional repositories and enhance the discoverability of academic resources. Both studies emphasized intelligent search and automated metadata extraction as key features for supporting efficient literature retrieval and reducing reliance on manual processes. However, the two studies differed in scope and application. While Smith and Anderson's research was conducted in a European university and primarily addressed general institutional knowledge management, the present study was implemented in SLTCFI and specifically focused on developing an AI-Integrated Research Archival System with additional features such as an AI-powered RRL Finder, cloud-based storage, mobile accessibility, and role-based user management. These enhancements extended the

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

functionality of the system beyond basic AI search and metadata extraction by tailoring it to the specific research needs of students, faculty, and administrators.

Chen et al. investigated the use of machine learning algorithms in digital libraries and concluded that AI-driven classification systems outperform traditional keyword-based retrieval methods. The study highlighted that automated tagging and semantic analysis improve the accuracy of search results, making digital archives more responsive to user needs in academic environments (Chen et al., 2021).

Chen et al.'s study was similar to the present research in that both utilized Artificial Intelligence to improve information retrieval in digital academic repositories. Both studies emphasized the importance of automated classification, tagging, and intelligent search mechanisms in enhancing the accuracy and relevance of search results for users. However, the two studies differed in their focus and system application. While Chen et al.'s research concentrated on digital libraries in a general academic context and evaluated machine learning algorithms for classification performance, the present study focused on the development of an AI-Integrated Research Archival System for SLTCFI, which integrated these AI techniques into a localized institutional repository with added features such as an AI-powered RRL Finder, cloud-based storage, mobile access, and role-based user management tailored to the specific needs of students, faculty, and administrators.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Khan and Abbas explored the adoption of cloud-based archival systems in higher education institutions and reported that cloud integration ensures scalability, data security, and continuous accessibility of research materials. Their findings support the idea that cloud-powered repositories are essential for long-term preservation and collaborative research management (Khan & Abbas, 2019).

Khan and Abbas's study was similar to the present research in that both emphasized the use of cloud technology to enhance the accessibility, scalability, and security of academic repositories. Both studies recognized cloud integration as a key factor in ensuring continuous access to research materials and supporting long-term data preservation. However, the two studies differed in terms of system features and technological focus. While Khan and Abbas primarily examined cloud-based archival systems for general research management in higher education institutions, the present study focused on the development of an AI-Integrated Research Archival System for SLTCFI that combined cloud storage with Artificial Intelligence features such as intelligent search, automated metadata tagging, and AI-powered RRL recommendations, thereby offering a more advanced and intelligent approach to research archiving and retrieval.

López and Martín analyzed the role of AI in academic knowledge management systems and found that intelligent recommendation engines significantly enhance literature discovery for students and researchers. The study emphasized that AI-

supported systems help users identify related studies more effectively, promoting research continuity and interdisciplinary learning (López & Martín, 2022).

López and Martín's study was similar to the present research in that both focused on the use of Artificial Intelligence to improve literature discovery and support research activities in academic environments. Both studies highlighted the importance of intelligent recommendation systems in helping users identify related studies more effectively and in promoting research continuity. However, the two studies differed in scope and implementation. While López and Martín examined AI-based knowledge management systems in a general academic context, the present study focused specifically on the development of an AI-Integrated Research Archival System for SLTCFI, which incorporated an AI-powered RRL Finder, cloud-based storage, and institutional-level user management to address the localized needs of students, faculty, and administrators.

Brown and Davis focused on digital preservation strategies in research repositories and stressed the importance of metadata standardization and automated backups. Their research demonstrated that institutions using structured digital archiving systems experienced fewer cases of data loss and improved reliability of academic records (Brown & Davis, 2018).

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Brown and Davis's study was similar to the present research in that both emphasized the importance of digital preservation strategies to ensure the reliability and long-term sustainability of academic records. Both studies highlighted the role of structured metadata and automated backups in preventing data loss and maintaining the integrity of research repositories. However, the two studies differed in their level of technological integration. While Brown and Davis primarily focused on general digital preservation practices and metadata standardization, the present study went further by developing an AI-Integrated Research Archival System for SLTCFI that incorporated AI-driven metadata generation, intelligent search, and automated RRL recommendations, thereby enhancing not only data preservation but also the efficiency and usability of research retrieval.

Nguyen et al. examined the integration of Optical Character Recognition (OCR) in academic repositories and found that OCR significantly improves access to older, scanned research documents. The study highlighted that digitization through OCR enables full-text search, making historical academic outputs more usable and relevant in modern research environments (Nguyen et al., 2020).

Nguyen et al.'s study was similar to the present research in that both emphasized the use of Optical Character Recognition (OCR) to improve access to archived academic documents, particularly older research materials in scanned formats. Both studies recognized that OCR-enabled digitization supports full-text

search and enhances the usability of historical academic outputs. However, the two studies differed in system scope and functionality. While Nguyen et al. focused mainly on the technical integration of OCR within academic repositories, the present study incorporated OCR as part of a broader AI-Integrated Research Archival System for SLTCFI that also included intelligent search, automated metadata tagging, AI-powered RRL recommendations, and cloud-based storage, thereby offering a more comprehensive and intelligent research management platform.

Rodriguez and Kim studied user acceptance of AI-powered library systems and concluded that proper user training and system transparency are critical for successful adoption. Their findings indicate that when users understand how AI supports research discovery, system usage and satisfaction significantly increase (Rodriguez & Kim, 2021).

Rodriguez and Kim's study was similar to the present research in that both emphasized the importance of user acceptance and proper system implementation in the success of AI-powered academic systems. Both studies recognized that user training and transparency are essential in ensuring that students and faculty effectively utilize AI features for research discovery. However, the two studies differed in application and system focus. While Rodriguez and Kim primarily examined user acceptance of AI-powered library systems in general, the present study applied these principles to the development of an AI-Integrated Research Archival System for SLTCFI

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

by incorporating user orientation modules, role-based access, and a user-friendly interface tailored specifically to the research needs of students, faculty, and administrators.

Patel and Sharma investigated ethical issues in AI-based information systems and emphasized the need for human oversight to ensure accuracy, fairness, and accountability. Their research supports a hybrid model where AI assists decision-making while humans validate outputs, particularly in academic and research contexts (Patel & Sharma, 2022).

Patel and Sharma's study was similar to the present research in that both emphasized the importance of balancing Artificial Intelligence with human involvement in academic information systems. Both studies supported a hybrid approach where AI assists in processing and decision-making while humans remain responsible for validating system outputs to ensure accuracy, fairness, and accountability. However, the two studies differed in terms of implementation and system context. While Patel and Sharma focused on ethical frameworks and general AI-based information systems, the present study applied these principles in a practical setting through the AI-Integrated Research Archival System for SLTCFI by integrating features such as an AI-powered RRL Finder with faculty and administrator verification, thereby maintaining scholarly oversight and research integrity within an institutional repository.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Wilson et al. explored the impact of real-time notification systems in digital repositories and found that automated alerts improve research engagement and monitoring of academic submissions. Their study showed that such features help faculty and administrators manage research workflows more effectively (Wilson et al., 2019).

Wilson et al.'s study was similar to the present research in that both recognized the importance of real-time notification features in enhancing research management and user engagement within digital repositories. Both studies emphasized that automated alerts support effective monitoring of academic submissions and improve communication among users. However, the two studies differed in system scope and integration. While Wilson et al. focused primarily on notification systems as a standalone feature in digital repositories, the present study incorporated real-time notifications as part of a broader AI-Integrated Research Archival System for SLTCFI, which combined this feature with AI-powered retrieval, automated metadata tagging, cloud-based storage, and role-based access to provide a more comprehensive and intelligent research management platform.

Garcia and Thompson analyzed modular system architecture in institutional repositories and concluded that flexible, scalable designs are essential for handling continuous growth of research data. Their study highlighted that modular systems

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

allow institutions to integrate new AI features without disrupting existing archival functions (Garcia & Thompson, 2023).

Garcia and Thompson's study was similar to the present research in that both emphasized the importance of flexible and scalable system architecture for managing institutional research repositories. Both studies recognized that modular designs enable the integration of new features, including AI functionalities, without disrupting existing archival operations. However, the two studies differed in scope and application. While Garcia and Thompson focused on the general principles of modular repository design, the present study applied these principles in developing the AI-Integrated Research Archival System for SLTCFI, which combined modular architecture with AI-powered RRL recommendations, automated metadata tagging, cloud-based storage, and user-specific access controls, ensuring both scalability and intelligent functionality tailored to the needs of students, faculty, and administrators.

Local

Sambrano et al. investigated the influence of Artificial Intelligence on library patronage among college students in Philippine higher education institutions, finding that AI tools were increasingly used for resource discovery and study support while complementing traditional library services, suggesting a hybrid model of human–AI

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

interaction for academic research environment (Sambrano, Reyes, Fabro, Colar & Medrano, 2025).

Sambrano et al.'s study was similar to the present research in that both examined the use of Artificial Intelligence to support academic research and enhance access to information in higher education settings. Both studies recognized that AI tools could complement traditional systems and improve resource discovery, promoting a more efficient and user-centered approach to managing academic materials. However, the two studies differed in scope and system design. While Sambrano et al. focused on general library patronage and the hybrid interaction between AI tools and students in Philippine universities, the present study specifically developed an AI-Integrated Research Archival System for SLTCFI, incorporating features such as an AI-powered RRL Finder, automated metadata tagging, cloud-based storage, and role-based access. This made the SLTCFI system not only a tool for discovery but also a comprehensive platform for managing, preserving, and retrieving institutional research outputs.

Factor, Manabat, Maghari & Balbas examined emerging trends in content management systems used by selected academic libraries in the Philippines, highlighting how web-based platforms enhance access to digital collections and support user engagement, demonstrating the growing role of digital systems in information management (Factor, Manabat, Maghari & Balbas, 2023).

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Factor, Manabat, Maghari, and Balbas' study was similar to the present research in that both emphasized the importance of digital systems in improving access to academic resources and supporting user engagement within higher education institutions. Both studies recognized that web-based platforms facilitate efficient management and retrieval of digital collections, enhancing the overall usability of academic repositories. However, the two studies differed in scope and functionality. While Factor et al. focused on general content management systems and trends in selected Philippine academic libraries, the present study specifically developed an AI-Integrated Research Archival System for SLTCFI, which went beyond basic content management by integrating AI-powered features such as intelligent search, automated metadata tagging, an AI-driven RRL Finder, and cloud-based storage, providing a more advanced, intelligent, and research-focused digital repository.

Lacuata explored the digitization of library resources in higher education institutions in La Union and found that while digitization guidelines were implemented, there were challenges in technical requirements and human resource planning, highlighting the need for stronger digital infrastructure and management strategies (Lacuata, 2021).

Lacuata's study was similar to the present research in that both addressed the digitization of academic resources to improve accessibility and preservation within higher education institutions. Both studies recognized that digital transformation

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

requires careful planning, infrastructure, and management to ensure effective implementation. However, the two studies differed in scope and technological integration. While Lacuata focused primarily on the challenges of implementing digitization guidelines in libraries in La Union, such as technical requirements and human resource planning, the present study went further by developing an AI-Integrated Research Archival System for SLTCFI, which combined digitization with AI-powered features like intelligent search, automated metadata tagging, and an AI-driven RRL Finder, along with cloud-based storage and role-based access, offering a more comprehensive and intelligent system tailored to the institution's research needs.

Alayon & David studied digital curation projects in selected academic and research repositories in the Philippines, identifying drivers, practices, and issues in digital preservation efforts, illustrating the local landscape of archiving practices and challenges in sustaining digital repositories (Alayon & David, 2016).

Alayon and David's study was similar to the present research in that both examined strategies for managing and preserving digital academic resources within Philippine higher education institutions. Both studies recognized the importance of structured practices and effective management to sustain digital repositories and ensure long-term accessibility of scholarly materials. However, the two studies differed in scope and system capabilities. While Alayon and David focused on identifying drivers, practices, and challenges in digital curation across selected repositories, the

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

present study specifically developed an AI-Integrated Research Archival System for SLTCFI, which not only addressed preservation but also integrated AI-driven features such as automated metadata tagging, intelligent search, an AI-powered RRL Finder, and cloud-based storage, thereby offering a more advanced, user-centered, and research-focused approach to digital archiving.

Gedulian examined the adaptability of librarians in selected academic libraries in Metro Manila regarding digital environments and found that librarians must continuously develop digital competencies to manage online resources effectively, underscoring the importance of user support and system literacy (Gedulian, 2025).

Gedulian's study was similar to the present research in that both recognized the critical role of user competence and digital literacy in the effective use of digital academic systems. Both studies highlighted that successful management and retrieval of online resources depend on well-trained personnel and user support mechanisms. However, the two studies differed in focus and application. While Gedulian concentrated on the adaptability and digital skill development of librarians in Metro Manila, the present study focused on the development of an AI-Integrated Research Archival System for SLTCFI, which not only provides a digital repository but also incorporates user orientation modules, a user-friendly interface, and AI-powered tools such as intelligent search and an RRL Finder to support students, faculty, and administrators in efficiently accessing and managing research outputs.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Mirador & Cajala investigated the digital transformation of academic libraries in the Philippines, revealing that institutions have made progress in implementing digital services and platforms but face ongoing challenges in fully integrating technologies that meet evolving user demands (Mirador & Cajala, 2024).

Mirador and Cajala's study was similar to the present research in that both focused on the digital transformation of academic libraries and the adoption of technology to improve access to resources. Both studies acknowledged that digital systems enhance service delivery and user engagement in higher education institutions. However, the two studies differed in scope and system design. While Mirador and Cajala primarily examined the general progress and challenges of implementing digital services in Philippine academic libraries, the present study specifically developed an AI-Integrated Research Archival System for SLTCFI, which not only digitized research outputs but also incorporated AI-driven features such as intelligent search, automated metadata tagging, an AI-powered RRL Finder, and cloud-based storage, thereby providing a more comprehensive and research-focused solution tailored to the needs of students, faculty, and administrators.

HERDIN PLUS (Health Research and Development Information Network) is the Philippines' national research information management system that digitally organizes and disseminates health research, showing a large-scale application of digital

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

repositories to centralize research outputs for institutional and national use (HERDIN PLUS, 2025).

HERDIN PLUS's system was similar to the present research in that both focused on centralizing academic and research outputs within a digital repository to improve accessibility, organization, and dissemination. Both emphasized the importance of structured digital systems in supporting research continuity and institutional knowledge management. However, the two systems differed in scope and application. While HERDIN PLUS addressed a large-scale, national-level repository primarily for health research, the present study focused on developing an AI-Integrated Research Archival System for SLTCFI, which targeted an institutional repository for diverse academic outputs. Additionally, the SLTCFI system integrated AI-driven functionalities such as intelligent search, automated metadata tagging, an AI-powered RRL Finder, and cloud-based storage, enhancing usability and supporting specific research needs of students, faculty, and administrators.

Aklatang Bayan Online was launched as a free online repository of Filipino books, journals, and creative writings, demonstrating an open-access digital collection initiative that enhances visibility and availability of scholarly materials in the Philippines (Aklatang Bayan Online, 2020).

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Aklatang Bayan Online was similar to the present research in that both aimed to enhance access to academic and scholarly materials through a centralized digital repository. Both emphasized improving the visibility and availability of resources to support research, learning, and knowledge sharing. However, the two initiatives differed in scope and functionality. While Aklatang Bayan Online primarily provided an open-access platform for Filipino books, journals, and creative writings, the present study focused on developing an AI-Integrated Research Archival System for SLTCFI, which combined digitization with AI-powered features such as intelligent search, automated metadata tagging, an AI-driven RRL Finder, cloud-based storage, and role-based access, providing a more interactive, intelligent, and research-focused platform for students, faculty, and administrators.

BAHANDIAN is the institutional repository of Central Philippine University, which curates and preserves scholarly work of faculty and students, exemplifying local efforts to manage and disseminate academic outputs digitally within a university context (BAHANDIAN, 2025).

BAHANDIAN was similar to the present research in that both focused on developing an institutional repository to digitally preserve and manage scholarly outputs of faculty and students. Both studies emphasized the importance of centralizing academic resources to improve accessibility, organization, and long-term preservation. However, the two systems differed in functionality and technological

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

integration. While BAHANDIAN primarily served as a digital repository for storing and disseminating academic work, the present study developed the AI-Integrated Research Archival System for SLTCFI, which extended these capabilities by incorporating AI-powered features such as intelligent search, automated metadata tagging, an AI-driven RRL Finder, cloud-based storage, and role-based access, thereby enhancing research efficiency, user experience, and intelligent retrieval of academic materials.

Digital Archives @ UP Diliman serves as the official institutional repository of the University of the Philippines Diliman, preserving theses, dissertations, and university records in digital format, illustrating a mature example of a local academic digital archiving system supporting long-term access and preservation (UP Diliman, 2025).

Digital Archives @ UP Diliman was similar to the present research in that both aimed to preserve and provide access to academic outputs through a centralized digital repository, ensuring long-term accessibility and organized management of scholarly materials. Both studies recognized the importance of systematic archiving in supporting research continuity and institutional knowledge. However, the two systems differed in scope and technological features. While Digital Archives @ UP Diliman primarily focused on storing theses, dissertations, and university records, the present study developed the AI-Integrated Research Archival System for SLTCFI, which expanded these functions by integrating AI-powered tools such as intelligent search,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

automated metadata tagging, an AI-driven RRL Finder, cloud-based storage, and role-based access, providing a more advanced, research-focused, and interactive platform tailored to the specific needs of students, faculty, and administrators.

Synthesis of the State of the Art

Studies by Smith & Anderson, Chen et al., and López & Martín highlighted the role of Artificial Intelligence in enhancing the discoverability of academic resources. Smith & Anderson demonstrated that intelligent search and automated metadata extraction significantly improved retrieval efficiency, reducing manual indexing efforts. Chen et al. emphasized that AI-driven classification and semantic tagging improved search accuracy in digital libraries. López & Martín showed that AI-powered recommendation engines helped users identify related studies more effectively, promoting research continuity. These insights collectively informed the development of the AI-Integrated Research Archival System for SLTCFI, particularly in designing the AI-powered RRL Finder and intelligent search features to improve efficiency, accuracy, and user experience.

Brown & Davis, Nguyen et al., and Alayon & David focused on strategies for digital preservation and metadata management. Brown & Davis emphasized structured metadata and automated backups to prevent data loss and ensure reliability. Nguyen et al. highlighted the importance of OCR in digitizing older research documents,

enabling full-text search and better accessibility. Alayon & David discussed drivers and challenges in digital curation within local repositories. These studies guided the SLTCFI system's implementation of automated metadata tagging, digitization of scanned research outputs, and structured repository design to ensure data integrity, usability, and long-term preservation.

Khan & Abbas and Garcia & Thompson explored cloud integration and modular system architectures. Khan & Abbas demonstrated that cloud-based systems ensure scalability, data security, and continuous accessibility of research materials, supporting long-term preservation. Garcia & Thompson highlighted that modular system design allows the seamless integration of new AI features without disrupting existing repository functions. These insights helped the present study incorporate cloud storage, modular architecture, and scalable AI functionalities in the SLTCFI system, ensuring both reliability and future expandability.

Patel & Sharma and Rodriguez & Kim emphasized the importance of balancing AI automation with human oversight. Patel & Sharma advocated for a hybrid approach in which AI assists decision-making while humans validate outputs to maintain accuracy, fairness, and accountability. Rodriguez & Kim highlighted the need for user training and transparency to ensure system adoption and effective use. These insights guided the SLTCFI system to implement faculty and administrator verification

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

alongside AI recommendations, ensuring ethical use, reliability, and informed user engagement.

Local studies by Sambrano et al., Gedulian, and Mirador & Cajala focused on user acceptance, digital competency, and engagement in digital repositories. Sambrano et al. noted the effectiveness of hybrid human–AI interactions in supporting resource discovery. Gedulian emphasized that librarians and users need continuous skill development to effectively utilize digital systems. Mirador & Cajala discussed challenges in fully integrating technologies to meet evolving user demands. These studies informed the design of the SLTCFI system's user interface, orientation modules, and training programs to facilitate smooth adoption and maximize usability for students, faculty, and administrators.

Local initiatives such as HERDIN PLUS, Aklatang Bayan Online, BAHANDIAN, and Digital Archives @ UP Diliman highlighted the value of institutional repositories in preserving and disseminating research outputs. HERDIN PLUS demonstrated centralized access to health research at a national level. Aklatang Bayan Online provided open-access collections to improve visibility. BAHANDIAN and UP Diliman's Digital Archives illustrated local efforts in long-term preservation of theses, dissertations, and scholarly outputs. These studies emphasized the importance of centralization, accessibility, and structured archiving, which influenced the SLTCFI

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

system's focus on preserving institutional research outputs and providing efficient retrieval through AI-enhanced features.

Gap Bridged by the Study

The most closely related studies to the present research, such as Smith & Anderson, Chen et al., and López & Martín, demonstrated the benefits of AI in improving information retrieval, metadata management, and literature discovery within institutional repositories. Similarly, local studies like BAHANDIAN, Digital Archives @ UP Diliman, and HERDIN PLUS highlighted the advantages of centralized digital repositories for preserving and providing access to academic outputs. While these studies provided important insights into digital archiving, AI integration, and user engagement, gaps remained in combining these elements within a single, institution-specific system that addresses the full spectrum of research management needs. Most prior systems either focused primarily on general digital preservation, search enhancement, or open-access dissemination, but did not integrate AI-driven tools with cloud storage, mobile accessibility, automated RRL generation, and role-based user management in a cohesive platform tailored to a local higher education context.

The present study addresses these gaps by developing the AI-Integrated Research Archival System for SLTCFI, which uniquely combines AI-powered functionalities, cloud-based storage, and user-centered features in a single platform.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Unlike previous systems, this study integrates an AI-driven RRL Finder for automated related literature recommendations, intelligent metadata tagging to reduce manual errors, real-time notifications for research updates, and mobile accessibility to support anytime, anywhere access. Additionally, role-based access ensures secure management of sensitive research outputs while maintaining faculty and administrator oversight, bridging the gap between AI automation and human validation. By merging digital preservation, intelligent retrieval, and localized institutional requirements, the present study offers a comprehensive, efficient, and context-sensitive research archival solution that was not fully realized in prior related studies.

Conceptual Framework

The conceptual framework of the study illustrated the systematic development of the Artificial Intelligence–Integrated Research Archival System through clearly defined inputs, processes, outputs, and feedback mechanisms. The input stage consisted of all technical components and system requirements necessary for building the platform. These inputs included the use of HTML, CSS, and JavaScript for front-end development, as well as an SQL-based database that supported structured storage and efficient retrieval of research records. Cloud storage and mobile accessibility were also incorporated to ensure secure and flexible research management. In addition, APIs were utilized to integrate external academic databases such as OpenAlex and Semantic Scholar, while Natural Language Processing (NLP) algorithms enhanced AI-driven

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

research recommendations. The system further required the development of key modules, including the AI-Based RRL Finder Module, Research Upload and Archiving Module, Role-Based Access Control Module, and Approval and Verification Module. Security and workflow optimization features were also defined at this stage, including user authentication, encrypted storage, secure cloud integration, and real-time notifications. The input phase concluded with the identification of evaluation criteria based on the ISO/IEC 25010 software quality model to assess functionality, efficiency, reliability, usability, maintainability, and portability.

The process stage followed the Agile Development Methodology, which ensured systematic, iterative, and user-centered development. Requirements analysis was conducted to identify user needs, institutional policies, and operational constraints. In the design phase, the system architecture, user interface layouts, data flow diagrams, and module integrations were formulated. This was followed by system development, where the identified modules, AI algorithms, and database structures were constructed and integrated. Testing and debugging were performed repeatedly throughout each development sprint to resolve errors, refine system behavior, and ensure alignment with specifications. After the system achieved functional stability, the release phase enabled deployment of the working version to selected stakeholders for evaluation and use. The maintenance stage followed, addressing post-release issues and integrating user feedback to enhance the system's performance and usability.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The output of the conceptual framework resulted in the development of the Artificial Intelligence–Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The completed system provided a centralized, secure, and intelligent platform for managing institutional research documents. It offered AI-powered literature recommendations, automated metadata generation, role-based access control, structured archiving, approval workflows, and real-time updates. These capabilities improved research retrieval, enhanced academic productivity, and supported institutional knowledge preservation. The system achieved its intended purpose of making research management more efficient, accessible, and streamlined for students, faculty, deans, administrators, and future researchers.

Finally, the framework incorporated a feedback mechanism wherein users—including students, faculty members, deans, and administrators—provided evaluations and recommendations after interacting with the system. Their feedback guided refinements, corrected functional limitations, and improved system responsiveness and usability. Continuous input from stakeholders ensured that the system evolved in accordance with academic needs, institutional policies, and technological advancements. This feedback loop reinforced the system's sustainability, scalability, and long-term relevance within SLTCFI.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

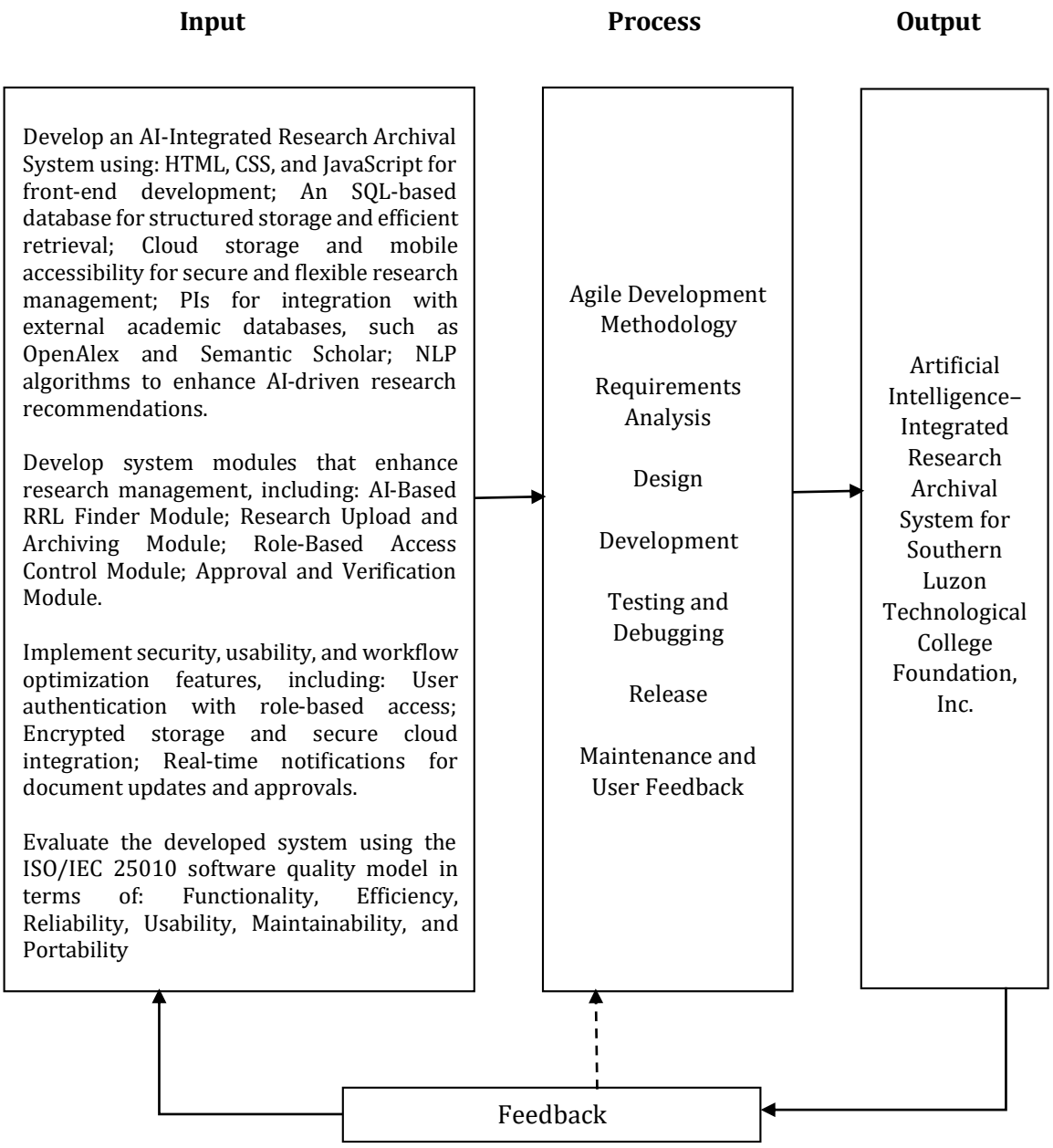


Figure 2.1

Conceptual Paradigm

CHAPTER III
TECHNICAL BACKGROUND

This chapter presents a discussion of the resources, stakeholders, constraints, and methodology utilized in developing the system.

Resources

The development of the Artificial Intelligence–Integrated School Research Archival System for Southern Luzon Technological College Foundation, Inc. (SLTCFI) requires reliable software and specialized development tools to ensure the system's functionality, accuracy, and security. These resources are grouped into software resources and development tools, which serve as the foundation for the successful design, implementation, and evaluation of the proposed system.

Table 3.1
Software Requirement

Particulars	Recommend Specifications
Operating System	Windows 10 or up
WebServer	Hostinger.ph
Database Manager	PhpmyAdmin
Programming Languages	HTML,PHP,MYSQL,JS,CSS
Framework	Bootstrap

Table 3.1 presents the software requirements essential for developing and deploying the AI-Integrated Research Archival System. The recommended operating

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

system for system development and testing was Windows 10 or later, ensuring compatibility with modern development tools and stable performance. The system utilized Hostinger.ph as the web server, providing reliable hosting services suited for web-based applications. PhpMyAdmin served as the database management tool, enabling efficient handling of SQL queries, database configuration, and structured data storage.

For system development, a combination of programming languages - including HTML, PHP, MySQL, JavaScript, and CSS - was employed to support both front-end interface design and back-end functionality. Bootstrap was used as the primary framework to enhance the system's responsiveness, user interface consistency, and mobile-friendly layout. Together, these software components ensured a functional, scalable, and accessible research archival platform.

Table 3.2

Hardware Requirement

Required Hardware	Recommended Hardware Specifications
Processor	13th Gen Intel(R) Core (TM) i5-13420H 2.10 GHz or Higher
Hard disk drive	500 GB or Higher
Memory	16.0 GB or Higher
Internet bandwidth	20 Mbps or higher

Table 3.2 outlines the hardware requirements necessary to ensure the smooth development, deployment, and operation of the AI-Integrated Research Archival

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

System. The recommended processor specification was a 13th Gen Intel® Core™ i5-13420H running at 2.10 GHz or higher, providing sufficient processing power for handling system development tasks, database operations, and AI-related functions. A minimum hard disk capacity of 500 GB was recommended to accommodate system files, research documents, backups, and other relevant data essential for system maintenance.

The system also required at least 16 GB of memory to support efficient multitasking, faster program execution, and stable operation of development tools and server processes. Additionally, an internet bandwidth of 20 Mbps or higher was recommended to ensure smooth access to cloud storage, API integrations, and seamless interaction with web-based system components. These hardware specifications collectively ensured reliable performance and optimized functionality during the system's development and actual use. Moreover, these requirements helped maintain system responsiveness even when handling large research files and simultaneous user activities. These specifications also facilitated a stable and efficient testing environment, allowing developers and evaluators to accurately assess system performance under realistic usage conditions.

Stakeholders

The success of the Artificial Intelligence–Integrated School Research Archival System for Southern Luzon Technological College Foundation, Inc. (SLTCFI)

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

depended on the active participation and collaboration of several key stakeholders. Each group played a crucial role in the design, development, evaluation, and implementation of the system, contributing to its overall effectiveness, usability, and alignment with the institution's objectives. Their coordinated involvement ensured that the system addressed actual user needs, supported academic functions, and adhered to institutional standards and requirements.

Students served as the primary end-users, uploading, accessing, and retrieving research papers, while benefiting from AI-driven features such as the Related Research Literature (RRL) Finder. Faculty members reviewed submissions, provided guidance, and verified content prior to dean approval. Deans acted as approvers, ensuring research quality and compliance with institutional standards, and they created faculty accounts to maintain user hierarchy. The school administrator managed overall system access, monitored activities, and ensured security and accountability. Future researchers and SLTCFI, as institutional stakeholders, also benefited from the system's archived research database, which supported long-term academic objectives and knowledge preservation.

In addition, IT staff provided technical support, maintained system infrastructure, and addressed any operational issues to guarantee smooth functionality. Curriculum coordinators advised on integrating research outputs into academic programs, ensuring alignment with learning outcomes. Alumni contributed

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

by sharing prior research work and professional insights, enriching the database for current students and faculty. External partners, including local industries and academic networks, also leveraged the system to foster collaborations and access a repository of scholarly resources.

Table 3.3

Stakeholders

Stakeholders	Task
Students	Serve as the primary end-users of the system who will upload, access, and retrieve research papers. They benefit from the system's AI-integrated features such as the Related Research Literature (RRL) Finder, which simplifies the process of locating relevant studies for their academic projects.
Faculty Members	Serve as reviewers who can view and download uploaded student research for evaluation, mentoring, and academic guidance. Faculty members also have the privilege to create student accounts, ensuring that each student researcher has secure access to the system. They assist in verifying the completeness of the student's submission prior to dean approval.
Dean	Act as approvers of research submissions within their respective departments. They review, validate, and approve uploaded research papers to ensure quality and compliance with institutional research standards before final archiving. Deans also have the authority to create faculty accounts, establishing the faculty-user hierarchy and ensuring proper account assignment.
School Administrator	Holds the highest administrative authority in the system and is responsible for creating and managing the dean accounts. The President oversees user role distribution, ensures that only authorized individuals gain system access, and monitors institutional-level system activity to maintain accountability.
Southern Luzon Technological College Foundation, Inc. (SLTCFI)	Serves as the principal institutional stakeholder and primary beneficiary of the system.

Table 3.3 presented the stakeholders involved in the development and implementation of the AI-Integrated Research Archival System, along with the specific tasks they performed throughout the process. Students served as the primary

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

end-users of the system, as they uploaded, accessed, and retrieved research papers. They benefited from the system's AI-powered features, including the Related Research Literature (RRL) Finder, which simplified the process of locating relevant studies for their academic requirements. Faculty members functioned as evaluators who viewed and downloaded student research for assessment, mentoring, and academic guidance. They were also responsible for creating student accounts and ensuring that submissions were complete prior to dean approval.

The deans acted as approvers of research outputs within their respective departments. They reviewed, validated, and approved uploaded research papers to ensure compliance with institutional research standards before documents were forwarded for final archiving. In addition, deans held the authority to create faculty accounts, thereby reinforcing the user hierarchy and ensuring accurate role assignment. The school administrator, serving as the highest authority in the system, managed the creation of dean accounts and oversaw user role distribution. The administrator ensured that only authorized personnel accessed the system and monitored overall institutional-level activity to maintain accountability and system integrity.

Finally, Southern Luzon Technological College Foundation, Inc. (SLTCFI) served as the principal institutional stakeholder and primary beneficiary, as the system supported its long-term goals of improving research management,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

strengthening academic processes, and enhancing institutional knowledge preservation.

Constraints

The development of the Artificial Intelligence–Integrated School Research Archival System for Southern Luzon Technological College Foundation, Inc. (SLTCFI) faced several constraints that influenced its design, implementation, and performance. One of the main limitations encountered was the dependency on stable internet connectivity and adequate server capacity, as the system's AI functions and data retrieval processes rely heavily on external sources. The integration of third-party academic databases such as OpenAlex and Semantic Scholar also presented constraints, as any service interruption, API modification, or access restriction from these external platforms could directly affect the accuracy and availability of related literature recommendations. Another constraint was the strict user hierarchy structure in which the President creates the Dean's account, the Dean creates the faculty's account, and the faculty creates the student's account. While this setup enhances system security and accountability, it may cause delays in user registration and account activation during the initial phase of system deployment.

Time

Time served as one of the major constraints in the development of the Artificial Intelligence–Integrated School Research Archival System for Southern

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Luzon Technological College Foundation, Inc. The entire research and system development process was allotted only six (6) months, which placed significant pressure on the researcher to complete each phase efficiently while ensuring the quality and functionality of the system. Given the extensive scope of the project - which involved requirements gathering, system design, interface creation, database structuring, AI integration, testing, evaluation, and documentation - the limited time frame required strict planning and disciplined execution.

To manage this constraint effectively, the researcher prepared a Gantt Chart outlining all major tasks and their corresponding timelines. This scheduling tool helped organize activities such as data gathering, module development, iterative testing, revision cycles, system deployment, and user evaluation. The chart also allowed the researcher to monitor progress, anticipate delays, and make timely adjustments to ensure adherence to the project schedule.

Despite the time limitation, the structured timeline and consistent monitoring enabled the researcher to complete all essential project components within the designated period. The Gantt chart thus played an essential role in maintaining workflow efficiency and ensuring that the system was developed, evaluated, and refined within the six-month development window.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Table 3.4

System Development Time Frame (Gantt chart)

Activities	Months					
	1	2	3	4	5	6
Requirements Analysis						
Design						
Development						
Testing and Debugging						
Release						
Maintenance and User Feedback						

Table 3.4 presented the Gantt chart that illustrated the time frame followed during the development of the Artificial Intelligence–Integrated School Research Archival System for Southern Luzon Technological College Foundation, Inc. The chart outlined the major activities carried out across the six-month development period, ensuring that each phase of the project was completed in an organized and timely manner.

The development process began with Requirements Analysis, which was conducted during the initial month. This phase involved gathering user needs, studying institutional workflows, and identifying system specifications that would guide the succeeding activities. Following this, the Design phase took place during the

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

second month, where system architecture, user interfaces, data flow diagrams, and database structures were prepared.

During the third and fourth months, the Development phase was implemented. In this period, the system's modules, functionalities, and AI components were coded, integrated, and refined. After development, Testing and Debugging occurred in the fourth and fifth months. This activity focused on identifying errors, improving system stability, ensuring compatibility, and verifying that all modules performed according to requirements.

The Release phase was carried out in the fifth month, during which the functional version of the system was deployed for institutional use and initial evaluation. Finally, in the sixth month, Maintenance and User Feedback took place. This stage involved collecting insights from users, conducting minor revisions, and ensuring that the system operated smoothly based on actual usage and identified concerns.

The Gantt chart served as a practical guide that helped the researcher monitor progress, allocate time efficiently, and ensure that each development phase was completed within the designated six-month period.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Financial

To ensure the continuous development and completion of the project, the researchers implemented a structured budgeting system. This financial planning process involved identifying all necessary resources, estimating costs, and allocating funds efficiently to maintain the balance between system performance and affordability. The use of a cost table simplified the budgeting process and provided transparency in the prioritization of expenditures. While most of the expenses were allotted to software tools, cloud services, and minimal hardware support, careful consideration was given to the functionality and performance each component contributed to the overall efficiency of the system. The researchers prioritized cost-effective solutions such as open-source development tools, free-tier cloud storage through Firebase, and publicly available AI APIs such as OpenAlex and Semantic Scholar to minimize financial impact without compromising system quality. The estimated project cost served as the financial framework to support development activities, testing, and documentation, ensuring that all required resources were acquired responsibly and efficiently within the academic timeframe. Additionally, the researchers ensured that all expenses remained aligned with institutional research guidelines and academic budgeting policies. This strategic financial planning helped prevent unnecessary spending and guaranteed that every allocated resource directly contributed to the system's successful completion.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Table 3.5

Estimated Cost of the Proposed System

Particulars	Cost (PHP)
Hardware (Development PC)	8,000
Server Hosting / Database Subscription	5,000
Testing & Evaluation	2,500
Documentation	3,000
Miscellaneous	1,500
TOTAL	**20,000**

Table 3.5 presents the estimated cost required for the development of the proposed Artificial Intelligence–Integrated Research Archival System. The financial allocation covered the essential resources needed to design, build, test, and document the system within the academic timeline. A total estimated budget of ₱20,000 was prepared to ensure that all necessary components were adequately supported.

A portion of the budget amounting to ₱8,000 was allotted for hardware, specifically for the development PC used throughout the system's coding, testing, and configuration processes. This ensured that the researcher had a reliable device capable of handling development tools and server simulations. Meanwhile, ₱5,000 was allocated for server hosting and database subscription, which covered the

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

deployment environment and cloud services needed to keep the system accessible and operational.

To ensure accuracy and functionality, ₱2,500 was assigned to testing and evaluation. This covered expenses such as debugging tools, test data preparation, and system performance validation. An allocation of ₱3,000 was dedicated to documentation, supporting the production of written reports, printing of materials, and preparation of supplementary documents required for academic submission. Finally, ₱1,500 was set aside for miscellaneous expenses, including unforeseen costs essential to maintaining workflow continuity.

The estimated budget of ₱20,000 provided a practical and cost-efficient financial framework that enabled the researcher to complete the project without compromising system quality, functionality, or academic requirements.

Project Development Methodology

The development of the Artificial Intelligence–Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. (SLTCFI) followed a systematic and iterative process to ensure accuracy, functionality, and user responsiveness. The project was carried out across several phases, each of which contributed to the refinement and completion of the system. These phases included

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Requirements Analysis, Design, Development, Testing and Debugging, Release, and Maintenance and User Feedback.

During the Requirements Analysis phase, the researcher gathered all essential user, technical, and institutional requirements needed for the system. Consultations with students, faculty members, deans, and administrators were conducted to identify the research workflow, access levels, data management needs, and AI-related functionalities expected of the system. This phase also involved reviewing existing research management practices in the institution to determine gaps that the system needed to address. All requirements were documented and prioritized to provide a clear foundation for development.

In the Design phase, the documented requirements were translated into system blueprints, interface layouts, data flow diagrams, and architectural plans. The researcher prepared the structural design of the database, outlined the modules of the system, and developed wireframes for the user interface. Special focus was given to designing the AI-Based RRL Finder, role-based access levels, and approval workflow. The system's navigation flow, cloud integration scheme, and API communication process were also fully designed. These design outputs served as references for the next phase.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The Development phase involved the actual construction of the system based on the established design. The front-end was developed using HTML, CSS, JavaScript, and Bootstrap, while the backend utilized PHP and an SQL-based database. The AI features, including NLP-based recommendation functions and external API integration with platforms like OpenAlex and Semantic Scholar, were programmed. Cloud storage and security mechanisms were implemented, and each system module-including user authentication, research upload, RRL Finder, and approval modules-was coded. The system evolved through multiple iterations as new functions were added and enhanced.

The Testing and Debugging phase ensured that all system components worked smoothly and correctly. The researcher conducted functionality tests, usability tests, compatibility checks, and database reliability tests. Errors, bugs, and performance issues were identified and corrected. Test users, including selected students and faculty members, were invited to evaluate the system and provide initial impressions. Their feedback allowed the developer to refine the interface, optimize loading speeds, and strengthen security functions. This phase ensured that the system met quality standards prior to deployment.

During the Release phase, the system was deployed to its hosting environment and made accessible to designated users. Login credentials were distributed, and the system's features were introduced to academic personnel. This phase included the

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

preparation of system documentation and user guides to support operational rollout. The system's performance was closely monitored to ensure stable operation and successful adoption within the institution.

Finally, the Maintenance and User Feedback phase focused on sustaining system performance after deployment. The researcher collected feedback from students, faculty members, and deans to identify usability issues, enhancement requests, or potential improvements. Minor updates were performed to improve workflow, fix newly discovered issues, and enhance overall user experience. This phase ensured that the system remained functional, updated, and aligned with institutional research needs.

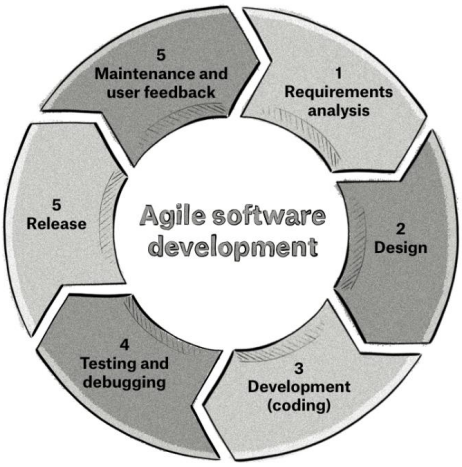


Figure 3.1
Agile Software Development

CHAPTER IV

ARTIFICIAL INTELLIGENCE-INTEGRATED SCHOOL RESEARCH ARCHIVAL SYSTEM FOR SOUTHERN LUZON TECHNOLOGICAL COLLEGE FOUNDATION

This Chapter covers the methodology process employed by the researcher in the development of the Artificial Intelligence-Integrated School Research Archival for Southern Luzon Technological College Foundation Inc. Using Agile Methodology

Requirements Analysis

The Requirements Analysis phase served as the foundation for the development of the Artificial Intelligence–Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. (SLTCFI). During this phase, the researcher gathered, examined, and validated all functional and non-functional requirements necessary to ensure that the system addressed the institution's research management needs. The process involved consultations, document reviews, and direct observations of existing research-handling procedures within SLTCFI. These activities allowed the researcher to identify gaps in the current manual and digital processes, particularly in areas related to research storage, accessibility, approval workflows, and literature retrieval.

The researcher conducted interviews and discussions with key stakeholders, including students, faculty members, deans, and administrators, to determine their

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

expectations, roles, and system-related concerns. Students emphasized the need for a more efficient way to upload research papers and locate related literature. Faculty members raised concerns about secure access, manageable review workflows, and consistent submission tracking. Deans highlighted the importance of standardizing research approval procedures, while administrators stressed the necessity of a centralized, secure, and scalable archival database. These inputs were carefully documented and translated into system features and requirements.

Technical requirements were also assessed during this phase. The researcher identified the need for AI-driven features such as the RRL Finder, automated metadata tagging, and content-based recommendation to support academic research tasks. Additionally, the analysis confirmed the importance of secure cloud storage, mobile-friendly access, and external API integration with platforms like OpenAlex and Semantic Scholar. Non-functional requirements, including system usability, reliability, performance efficiency, data security, and compliance with institutional standards, were also outlined to ensure adherence to the ISO/IEC 25010 quality model.

The Requirements Analysis phase provided a comprehensive understanding of user needs, institutional workflows, and technical constraints. The insights gathered during this stage served as the blueprint for the system's design and development, ensuring that every component of the AI-Integrated Research Archival

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

System aligned with the intended objectives and addressed the operational challenges of SLTCFI's research environment.

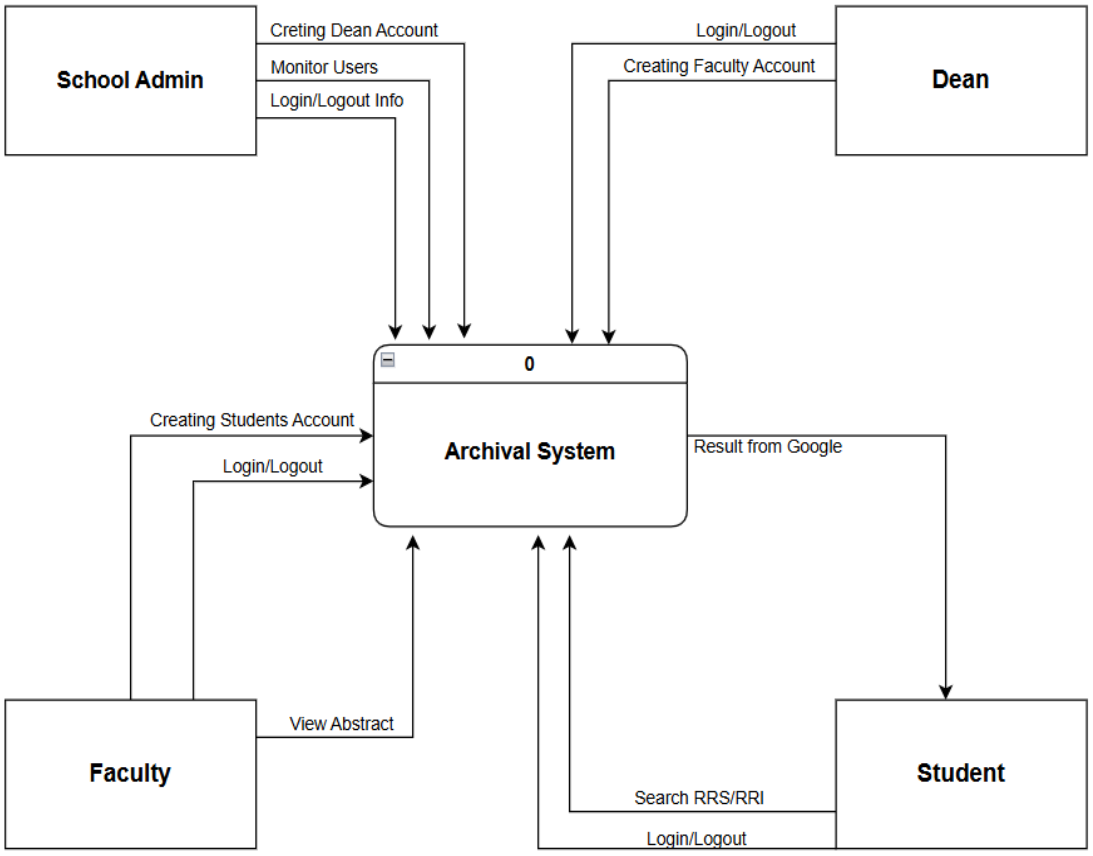


Figure 4.1

Data Flow Diagram of the Existing System

Figure 4.1 illustrated the Data Flow Diagram (DFD) of the existing system, showing how research documents were manually processed, reviewed, and stored within SLTCFI. The diagram depicted the traditional workflow in which students

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

submitted printed or digital copies of their research papers directly to faculty members or academic offices, where submissions were reviewed, approved, and archived through separate and often uncoordinated steps. It also showed how faculty and deans relied on manual verification, email exchanges, and physical storage to manage research outputs, leading to delays, inconsistencies, and potential risks in document handling. By presenting the movement of data across users, processes, and storage points, the DFD highlighted the inefficiencies and bottlenecks of the current system and provided a clear basis for identifying areas that needed improvement through automation and AI integration.

Design

The Design section discussed how the proposed Artificial Intelligence–Integrated Research Archival System was structured and organized based on the requirements identified in the analysis phase, and it also presented the complete set of system designs that guided the development process. This section explained the architectural layout of the system, including the user interface design, database schema, security components, and workflow structure. It further detailed how each module - such as the AI-Based RRL Finder, Research Upload and Archiving Module, Role-Based Access Control Module, and Approval and Verification Module - was designed to address the gaps identified in the existing system's Data Flow Diagram. In addition, the section included diagrams such as the system architecture, entity–

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

relationship model, and enhanced DFDs, all of which illustrated how data would flow more securely, efficiently, and intelligently within the new automated system. Overall, the design section served as the comprehensive blueprint that outlined the system's structure, behavior, and integration components, ensuring a clear foundation for the subsequent development phase.

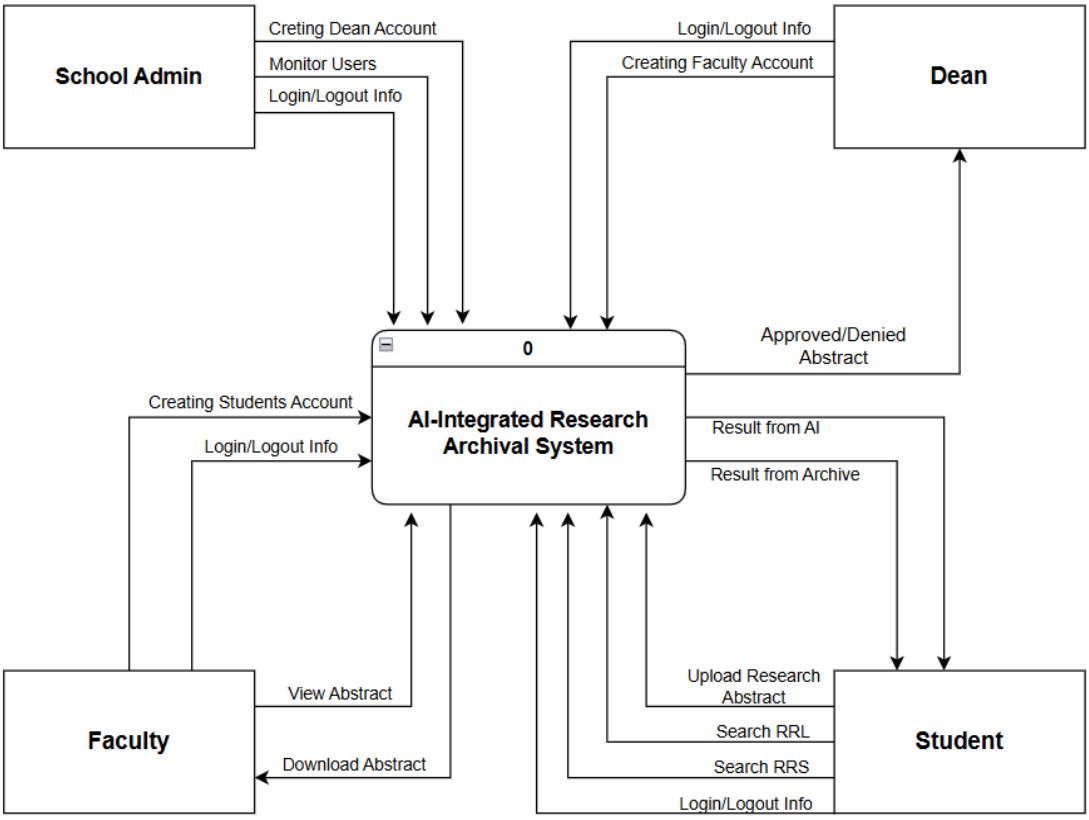


Figure 4.2

Data Flow Diagram of the Developed System

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The Data Flow Diagram (DFD) of the Developed System, shown in Figure 4.2, illustrated how information flowed within the newly created Artificial Intelligence–Integrated Research Archival System for SLTCFI. Unlike the existing manual process, the developed DFD presented a more structured, automated, and secure flow of data among users, system modules, and the database. It depicted how students uploaded research documents, how the AI module processed metadata and generated literature recommendations, and how faculty and deans reviewed and approved submissions through interconnected processes. The diagram also showed the role-based access paths, ensuring that each user-student, faculty, dean, or administrator--interacted only with authorized system functions.

Furthermore, the developed DFD highlighted the integration of external APIs such as OpenAlex and Semantic Scholar, illustrating how the system retrieved supplementary academic information to enhance the RRL Finder feature. It also mapped the secure storage and retrieval of research files through cloud integration, demonstrating improved workflow efficiency compared to the traditional process. Figure 4.2 provided a clear visualization of the enhanced data movement, system automation, and intelligent processing capabilities that addressed the limitations of the existing research management workflow.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

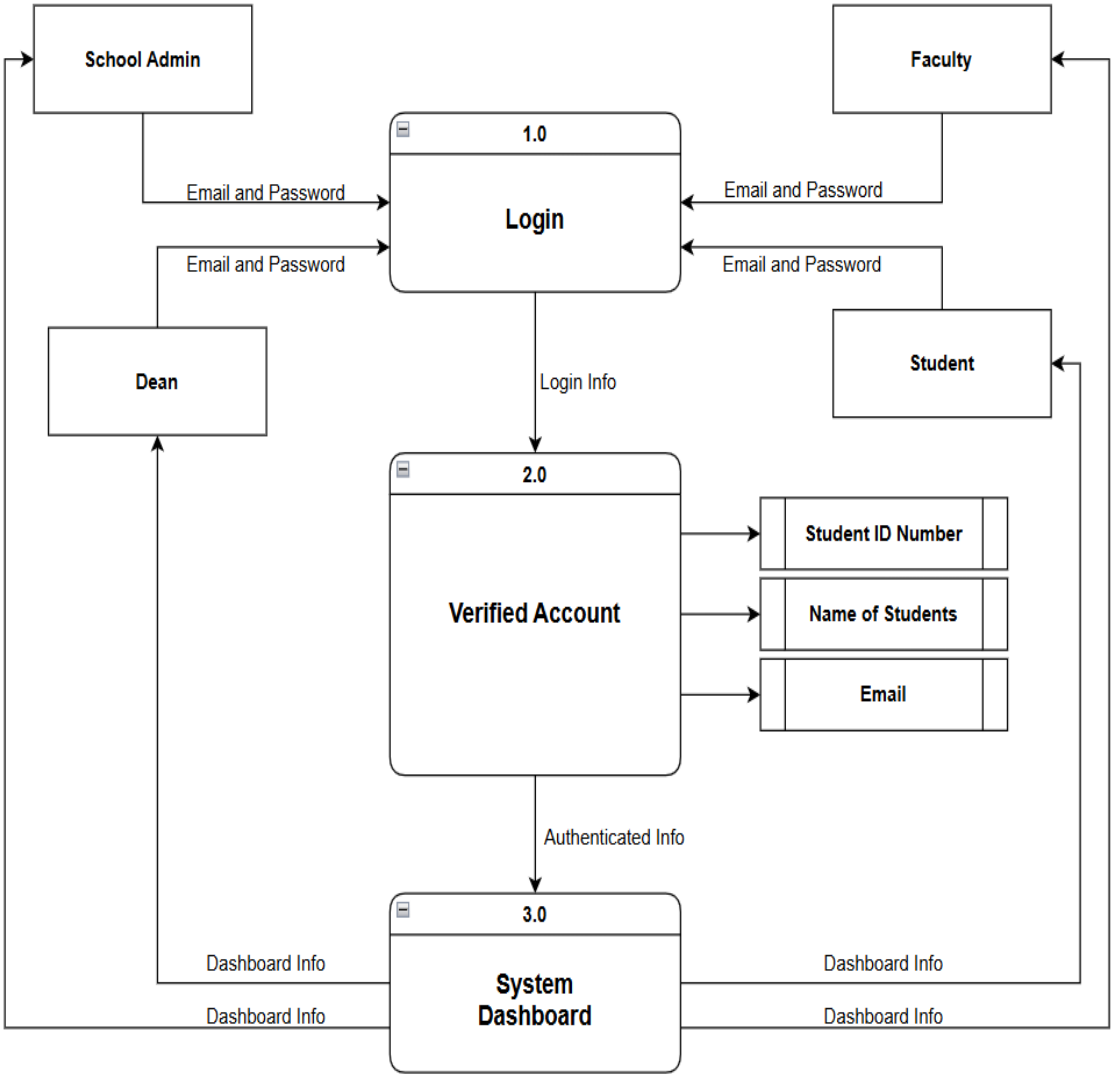


Figure 4.3

Data Flow Diagram of the Login Module

The Data Flow Diagram (DFD) of the Login Module, shown in Figure 4.3, illustrated the authentication process that controlled access to the developed AI-Integrated Research Archival System. The diagram showed how users - whether

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

students, faculty, deans, or administrators - entered their credentials, which were then transmitted to the system for validation. The DFD depicted the flow of input data from the login interface to the authentication process, where the system cross-checked the provided username and password against the records stored in the database. If the credentials matched, the user was granted access to their designated role-based dashboard; if not, the system returned an error message prompting re-entry.

The diagram highlighted the importance of security in the login process, showing how encrypted verification protected user information during authentication. The DFD emphasized not only the validation of credentials but also the assignment of user roles, which determined what specific modules or functions each user could access within the system. Through this process flow, Figure 4.3 demonstrated how the login module served as the first layer of security, ensuring that only authorized users were allowed to interact with the system's research management features.

Additionally, the DFD showed how failed login attempts were logged for monitoring and auditing purposes, supporting system accountability and security oversight. The flow diagram also illustrated the interaction between the login module and the central database, highlighting how authentication data was securely stored and retrieved in real time. Finally, the diagram underscored the modularity of the

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

login process, allowing for future enhancements such as multi-factor authentication or integration with external identity providers without disrupting the existing system architecture.

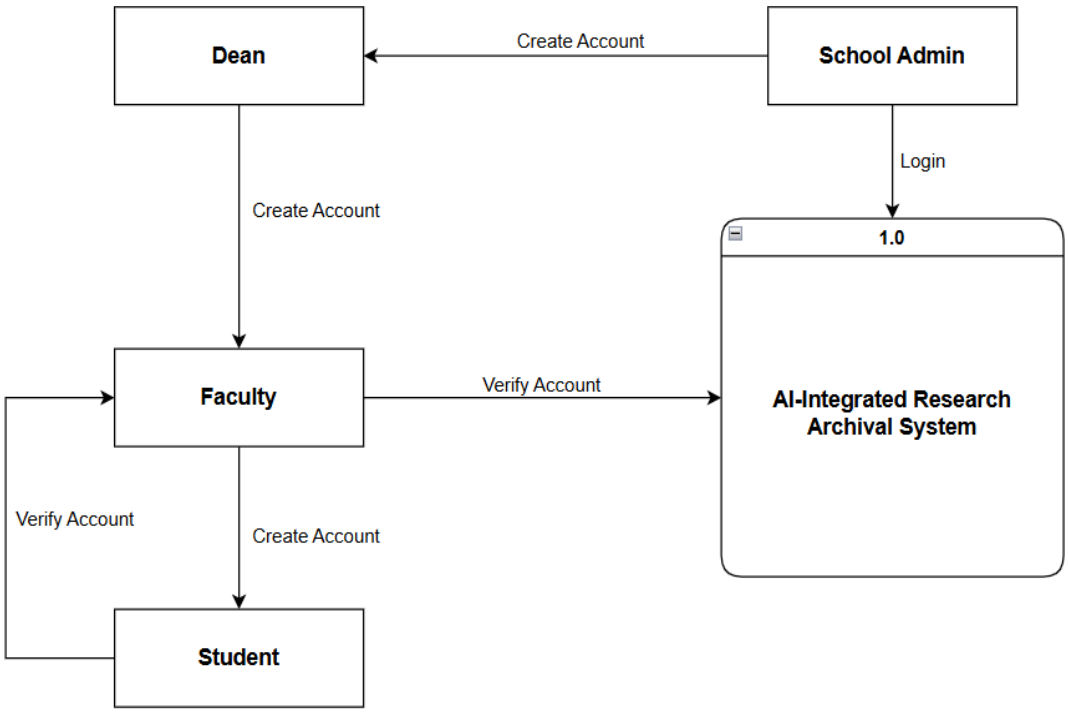


Figure 4.4

Creating Account Module

The Data Flow Diagram (DFD) of the Creating Account Module, shown in Figure 4.4, illustrated how new user accounts were generated and securely stored within the developed AI-Integrated Research Archival System. The diagram showed the process beginning with the authorized user - typically a faculty member or dean - who entered the required user information through the account creation interface. This information, which included the user's full name, role, username, and temporary

password, was then transmitted to the system for validation and processing. The system ensured that only authenticated personnel with assigned privileges could perform this task, preventing unauthorized creation of user accounts. This verification step helped maintain the integrity of the institutional user hierarchy.

The DFD demonstrated how the system checked the submitted data for completeness, duplication, and validity before allowing it to proceed to the next step. Once validated, the information was sent to the database, where a new account record was created and securely stored. The diagram also showed the system's feedback process, where a confirmation message was returned to the account creator, indicating that the new user account had been successfully generated. If errors or missing data were detected, the system returned a notification prompting corrections.

Figure 4.4 emphasized the controlled and role-based nature of account creation within the system. By illustrating how only authorized personnel could generate new accounts and how the system ensured accuracy and security during the process t- he diagram highlighted the importance of maintaining a structured and secure user management workflow in supporting the institution's research operations. Additionally, the DFD captured the logging of account creation activities, allowing administrators to track and audit user management actions for accountability. The diagram also suggested the system's ability to handle future

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

updates or modifications to user roles without compromising existing security protocols.

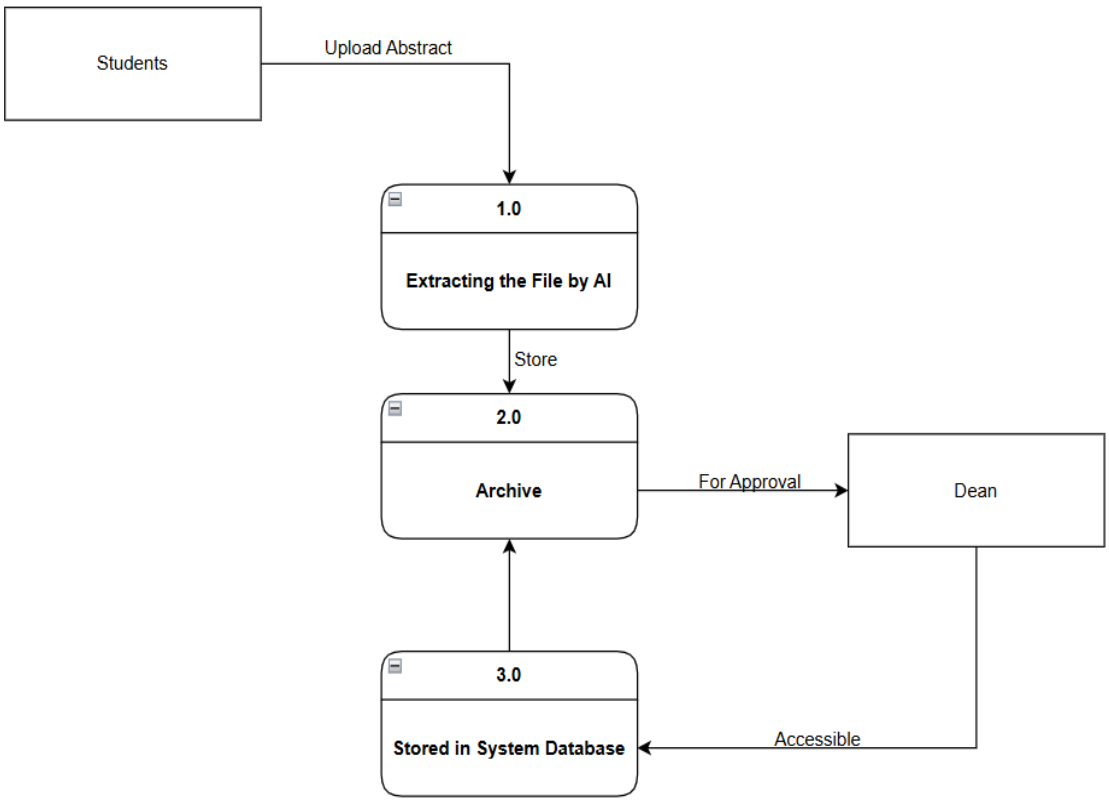


Figure 4.5

Data Flow Diagram of the Research Abstract Uploading Module

The Data Flow Diagram of the Research Abstract Uploading Module, shown in Figure 4.5, illustrated how research abstracts were submitted, processed, and stored within the developed AI-Integrated Research Archival System. The diagram showed that the process began when a student or authorized user uploaded the research

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

abstract through the designated upload interface. The system first received the file and accompanying metadata - such as title, author, department, and keywords - which were required to properly classify and index the research entry.

The DFD demonstrated how the system validated the uploaded abstract by checking file format, completeness of required fields, and potential duplication within the database. If the file met the validation criteria, the system proceeded to store the abstract securely in the cloud-based repository while simultaneously recording its metadata in the SQL database for easy retrieval and indexing. The diagram also depicted the workflow path leading to the dean of faculty review queue, where the abstract awaited verification and approval before final archiving.

If any errors were detected - such as unsupported file format, missing details, or upload failure - the system provided immediate feedback to the user, prompting them to correct and resubmit the file. Once successfully uploaded and stored, the system generated a confirmation message, ensuring transparency and completion of the process. The DFD also illustrated the time-stamping of each submission, enabling accurate tracking of submission dates and review timelines. Furthermore, it showed how the system maintained version control to prevent data inconsistencies and ensure that only the most recent, approved abstract was archived.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.5 highlighted the streamlined and secure workflow for research abstract submission, emphasizing validation, proper documentation, organized storage, and seamless transition to the approval process. This ensured that every uploaded abstract was accurately recorded, classified, and prepared for institutional review and archiving.

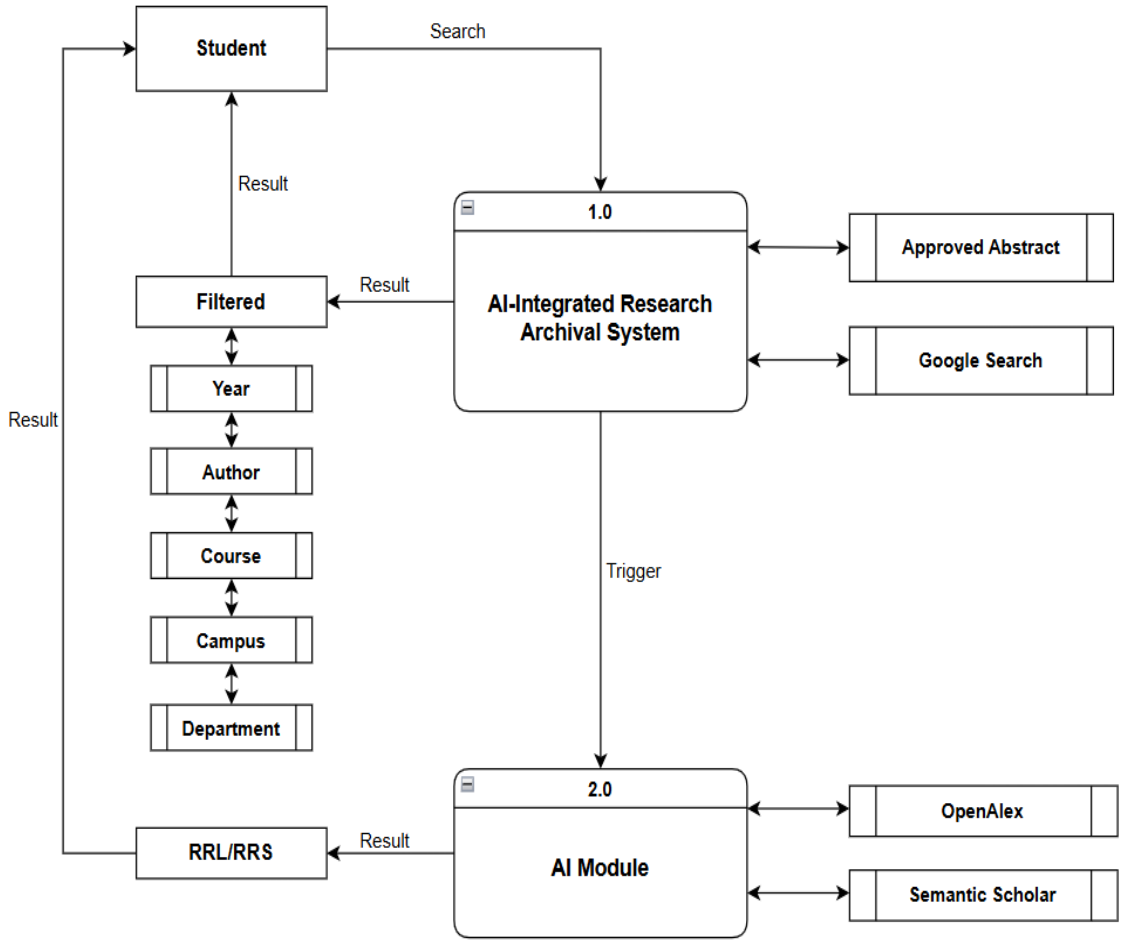


Figure 4.6

Data Flow Diagram of the Research Abstract Uploading Module

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The Data Flow Diagram of the Research Abstract Uploading Module, shown in Figure 4.6, illustrated the complete process by which research abstracts were submitted, validated, stored, and routed for approval within the developed AI-Integrated Research Archival System. The workflow began with the student or authorized user who initiated the upload by selecting the research abstract file and providing essential metadata such as the research title, author details, keywords, and academic program. This initial step ensured that each entry was properly identified and classified before entering the system.

The diagram showed how the system performed an automatic validation check on both the file and its metadata. This validation process included verifying the file format, checking for missing or incomplete information, examining possible duplication of previously submitted abstracts, and ensuring the integrity of the uploaded document. Once validated, the system proceeded to store the research abstract securely in the cloud repository, while simultaneously saving its metadata into the SQL database for indexing, searching, and retrieval. The DFD also illustrated how the uploaded abstract was automatically forwarded to the designated faculty reviewer or dean for further verification and approval, initiating the academic review workflow.

If the submitted file failed any validation criteria - for example, due to incomplete metadata, an unsupported file type, or upload errors - the system

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

immediately returned an alert to the user, prompting them to revise and re-upload the document. Once a submission successfully passed validation and storage, the system generated a confirmation message, ensuring transparency and notifying the user that the research abstract had been successfully recorded.

Figure 4.6 emphasized the importance of accuracy, security, and structured workflow in handling research submissions. By presenting each step - from upload to

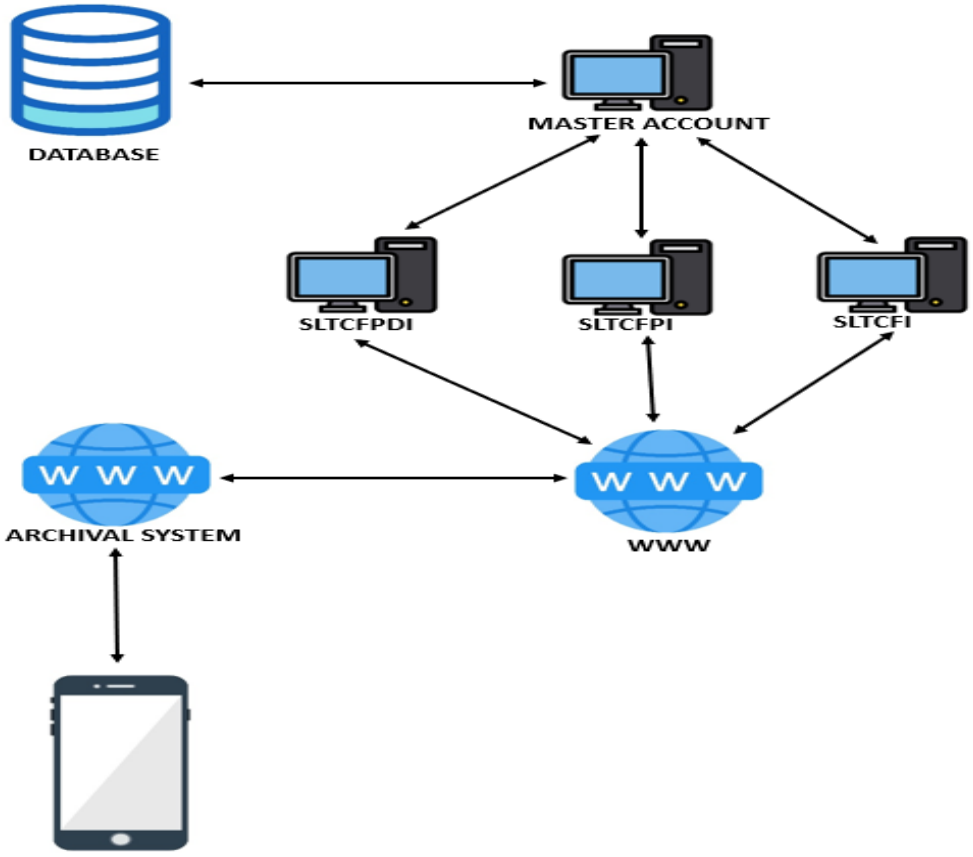


Figure 4.7

System Architecture of the Developed System

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The System Architecture of the Developed System, shown in Figure 4.7, illustrated the overall structural design and interaction of components within the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The architecture depicted a multi-tiered framework that included the user interface, application logic, database, cloud services, and external API integrations. The diagram showed how users - students, faculty, deans, and administrators - interacted with the system through the front-end interface developed using HTML, CSS, JavaScript, and Bootstrap, ensuring a responsive and intuitive user experience.

The architecture further highlighted the backend application layer, which handled core functionalities such as AI-powered RRL Finder operations, research uploading and archiving, role-based access control, and approval workflows. The backend was supported by an SQL-based database that managed structured storage, indexing, and retrieval of research documents. Cloud storage integration was incorporated to provide secure, scalable, and mobile-accessible storage for research files, while external APIs such as OpenAlex and Semantic Scholar enabled the system to fetch additional academic resources for literature recommendations.

Figure 4.7 illustrated the security mechanisms embedded within the architecture, including user authentication, encrypted data storage, and role-based permissions. The diagram also showed the system's communication flow, where data

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

moved seamlessly between the front-end, application logic, database, and external services, ensuring efficiency, reliability, and integrity of research information. Overall, the system architecture provided a visual representation of how all modules and components were organized.

Development

The Development Phase involved the actual creation and programming of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. (SLTCFI). During this phase, the researcher translated the system design into functional modules using front-end technologies such as HTML, CSS, JavaScript, and Bootstrap, along with backend tools like PHP and an SQL-based database. Cloud integration and external APIs, including OpenAlex and Semantic Scholar, were incorporated to enhance AI-driven recommendations and ensure mobile accessibility.

This phase also included the development of key system modules, each of which was captured in a series of screenshots to illustrate functionality, layout, and user interaction. The Login Module screenshot demonstrated secure authentication and role-based access, ensuring that only authorized users could enter the system. The Creating Account Module screenshots showed how administrators and faculty could generate new user accounts with proper validation and secure storage.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Screenshots of the Research Uploading Module illustrated the step-by-step process of submitting abstracts, attaching metadata, and automatically routing submissions for faculty and dean review. Similarly, the AI-Based RRL Finder Module screenshots highlighted how the system suggested related literature based on keywords and content analysis, demonstrating the AI functionality in supporting research management.

Additional screenshots captured the Approval and Verification Module, which displayed the interface where faculty and deans reviewed, validated, and approved submissions. The series also included representations of real-time notifications, cloud storage access, and role-specific dashboards, showing how data flowed across modules and how users interacted with the system in practical scenarios. Collectively, these screenshots provided visual evidence of the system's functionality, design implementation, and integration of AI features, confirming that the developed modules aligned with the requirements identified during the analysis and design phases.

The developed AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., is shown in the screenshots below:

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

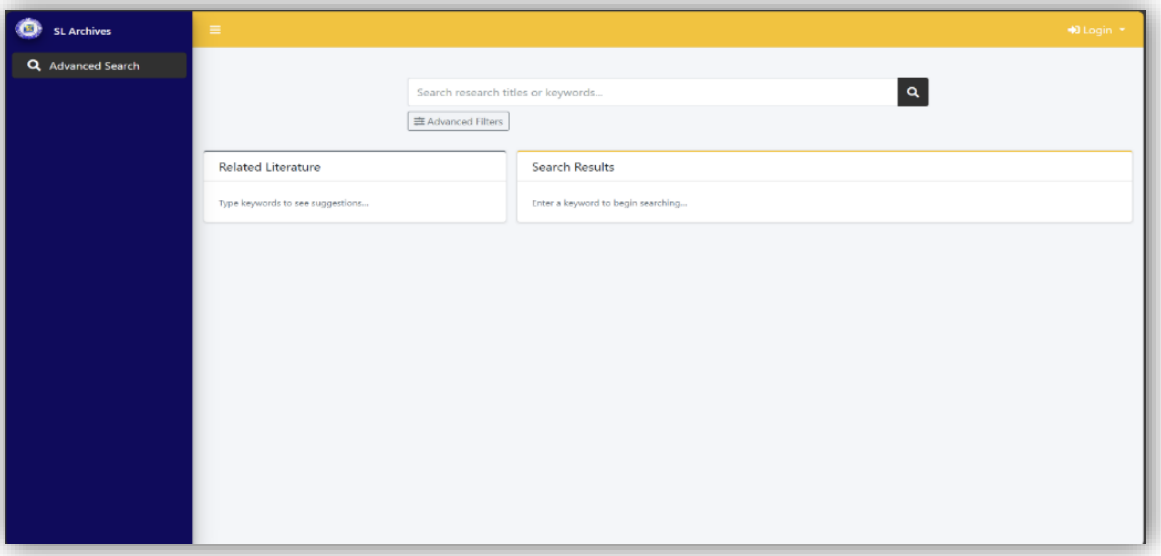


Figure 4.8

Home Page

Figure 4.8 illustrated the Home Page of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., serving as the main gateway for all users. The screenshot displayed a clean and user-friendly interface that allowed students, faculty, deans, and administrators to navigate the system efficiently. Key elements, such as login access, quick links to system modules, notifications, and announcements, were prominently featured to facilitate immediate interaction. The home page also highlighted role-based accessibility, ensuring that users only saw options and functions relevant to their assigned permissions. Overall, Figure 4.8 demonstrated how the home page provided an organized, intuitive, and

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

visually appealing entry point, reflecting the system's design emphasis on usability, accessibility, and seamless navigation.

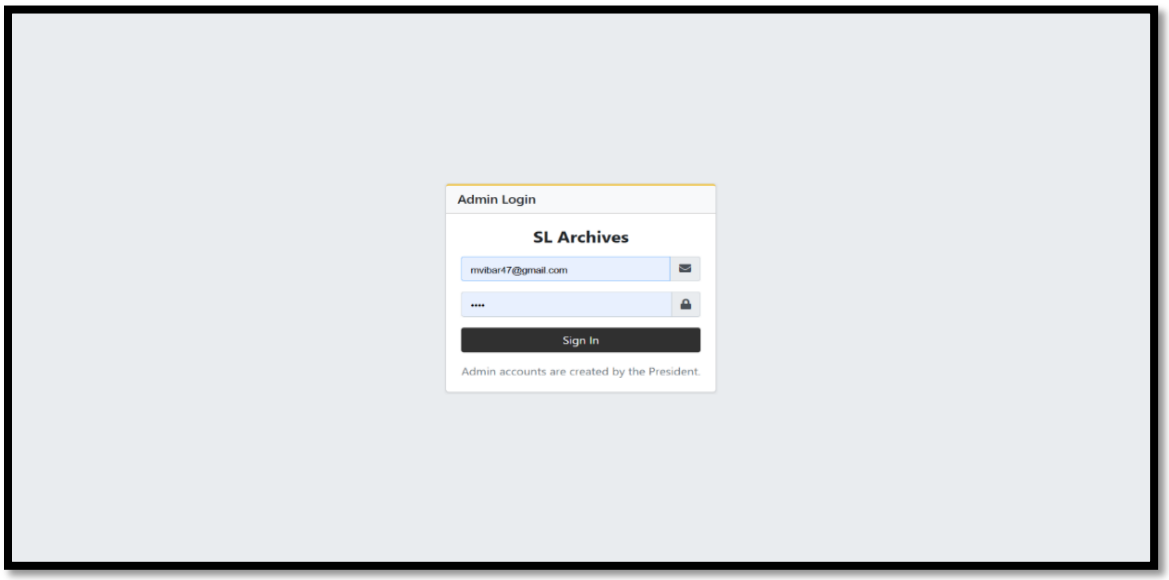


Figure 4.9

Login Module

Figure 4.9 illustrated the Login Module of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which served as the secure entry point for all users. The screenshot displayed the login interface where users - students, faculty, deans, and administrators - entered their credentials, including username and password, to access the system. It highlighted the system's role-based authentication, ensuring that users were granted permissions according to their assigned roles. Additionally, the module incorporated security measures such as encrypted data transmission and validation checks to prevent unauthorized access. Figure 4.9 demonstrated how the login module provided a secure, user-

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

friendly, and efficient gateway that protected sensitive research data while supporting seamless access for authorized users.

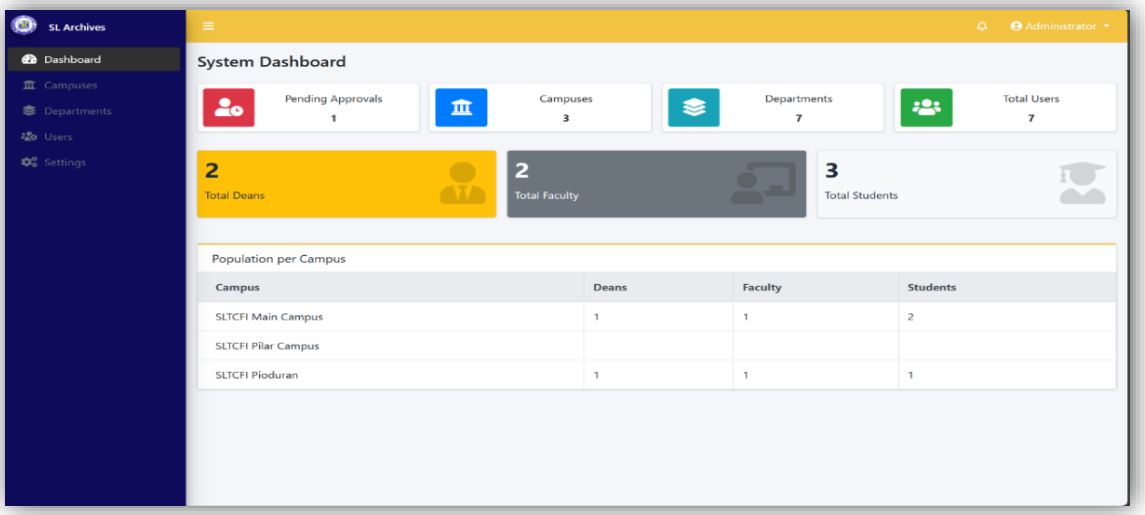


Figure 4.10

The Administrator System Dashboard

Figure 4.10 illustrated the Administrator System Dashboard of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., serving as the central control panel for system management. The screenshot displayed an organized interface that allowed administrators to oversee user accounts, monitor system activity, manage roles, and track research submissions across all departments. Key features, such as user account creation, role assignment, system notifications, and access logs, were prominently presented to facilitate efficient administrative control. The dashboard also provided quick navigation to

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

core modules and reports, enabling administrators to make informed decisions and maintain system security.

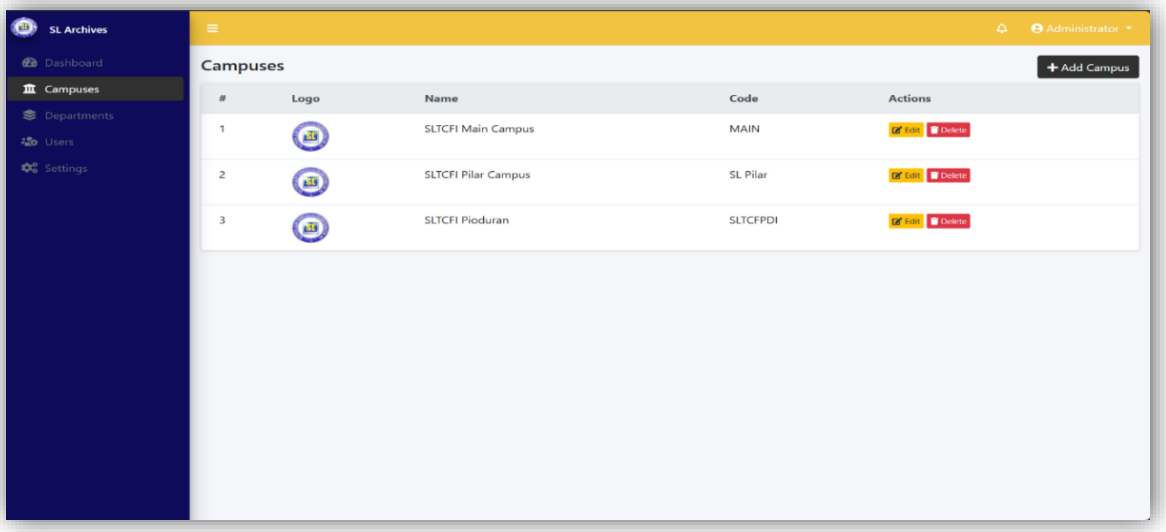


Figure 4.11

The Administrator List of Campuses Tab

Figure 4.11 illustrated the Administrator "List of Campuses" Tab within the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which allowed administrators to manage and oversee multiple campuses or departments. The screenshot displayed a structured interface where campus names, codes, and related details were listed, enabling administrators to add, update, or remove campus records as needed. The tab provided quick access to associated faculty, students, and research submissions for each campus, ensuring that institutional data remained organized and easily navigable. Figure 4.11 demonstrated how this feature supported efficient campus-level management,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

streamlined user oversight, and enhanced the overall administrative functionality of the system.

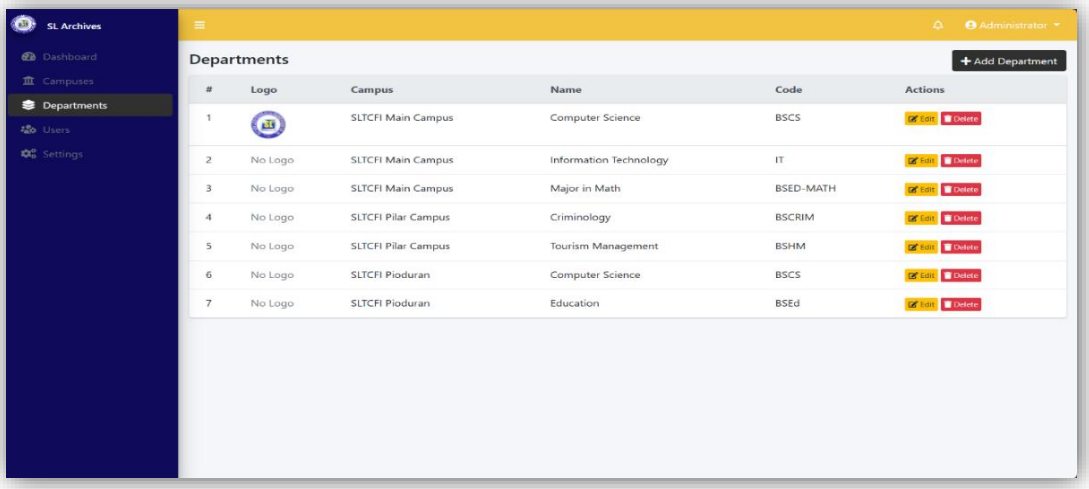


Figure 4.12

List of Departments Tab

Figure 4.12 illustrated the Administrator "List of Departments" Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which enabled administrators to manage and organize all academic departments within the institution. The screenshot displayed a structured table showing department names, codes, and associated details, allowing administrators to add, edit, or remove department records as needed. This tab also provided quick access to faculty and student assignments within each department, as well as oversight of research submissions and module interactions. Overall, Figure 4.12 demonstrated how the feature supported efficient department-level management,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

ensured accurate data organization, and enhanced the administrative control and functionality of the system.

Additionally, the tab incorporated search and filtering functions, allowing administrators to quickly locate specific departments and review their associated records. It also ensured that any updates made to department information were automatically synchronized across related modules, maintaining consistency and integrity within the system database.

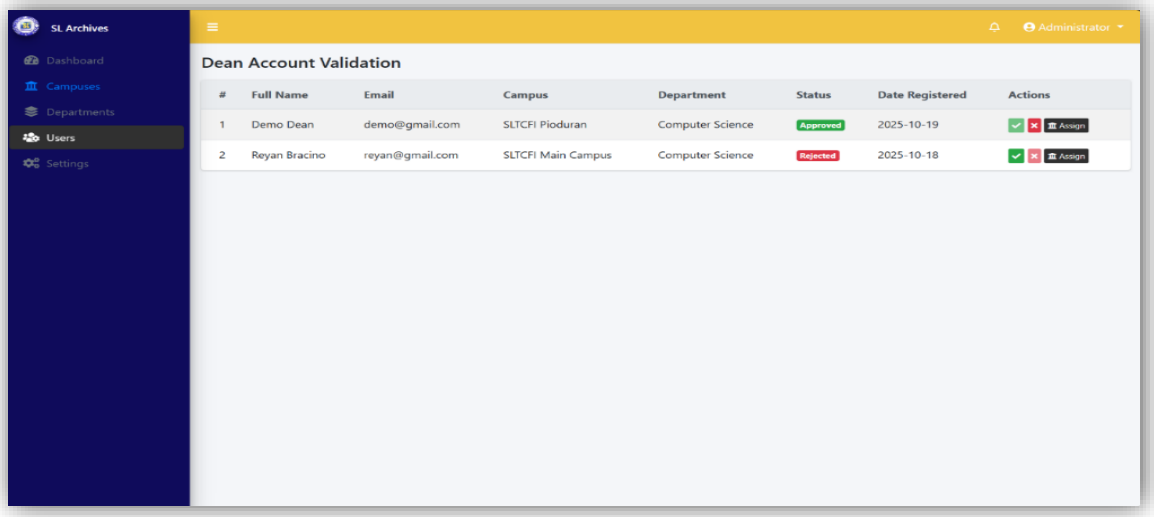


Figure 4.13
Account Validation Tab

Figure 4.13 illustrated the Administrator "Account Validation" Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which allowed administrators to review and approve new user

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

accounts. The screenshot displayed a list of pending accounts with relevant details, including usernames, roles, and registration dates, providing administrators with the ability to verify the authenticity and completeness of each submission. The tab also included options to approve, reject, or request additional information for account entries, ensuring that only authorized users gained access to the system. Figure 4.13 demonstrated how the account validation features enhanced security, maintained role-based access control, and supported proper management of user authentication within the system.

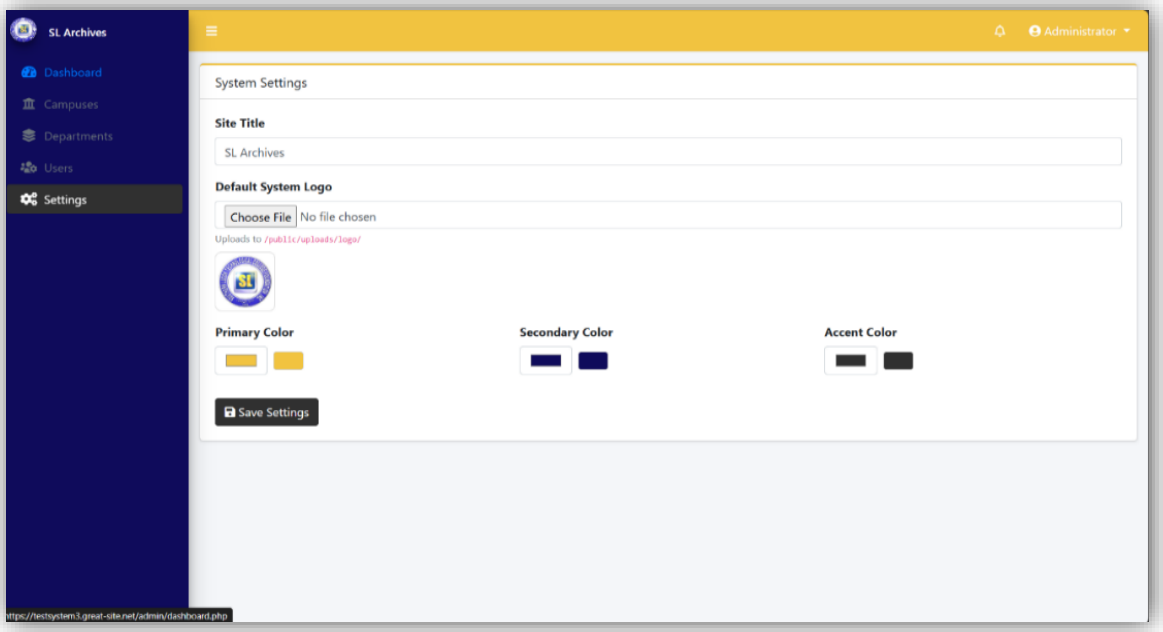


Figure 4.14
System Settings Tab

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.14 illustrated the Administrator "System Settings" Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which provided administrators with tools to configure and customize system parameters. The screenshot displayed options for adjusting security settings, managing user roles, setting system preferences, and configuring notification protocols. It also allowed administrators to update database connections, control module accessibility, and oversee system-wide operations to ensure smooth and secure functionality. Overall, Figure 4.14 demonstrated how the System Settings tab empowered administrators to maintain system integrity, enforce policies, and optimize overall performance within the research archival platform.

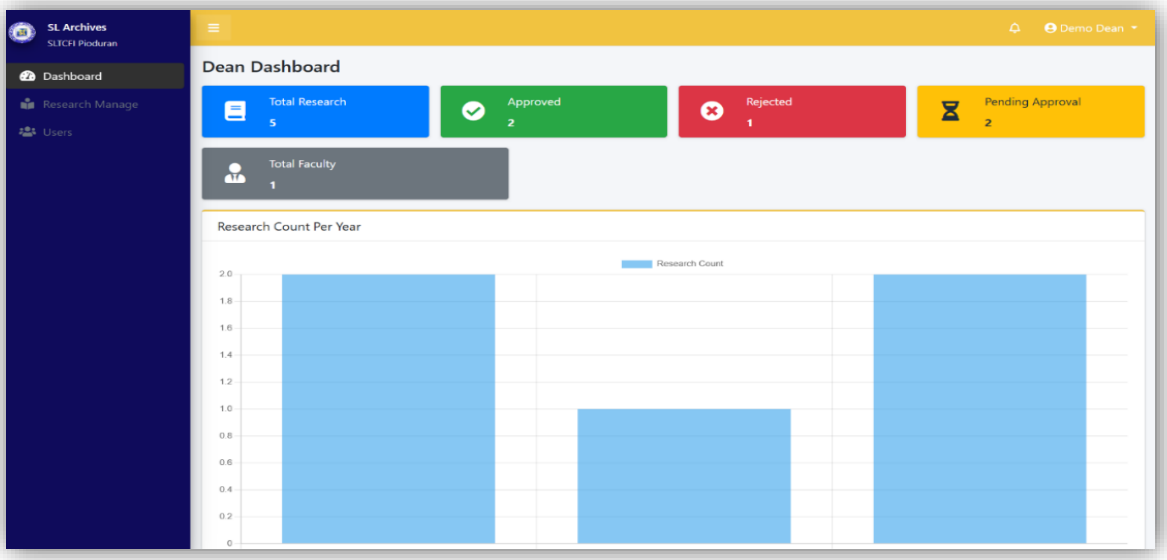


Figure 4.15

The Dean's Dashboard

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.15 illustrated the Dean's Dashboard of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which served as the central interface for department-level academic oversight. The screenshot displayed an organized view of pending research submissions, approval status, notifications, and faculty activities, enabling deans to efficiently monitor and manage the research workflow within their departments. Key features, such as review queues, submission tracking, and role-specific access controls, were highlighted to facilitate timely evaluation and approval of student research outputs. Figure 4.15 demonstrated how the Dean's Dashboard provided a user-friendly, efficient, and secure interface for academic supervision and decision-making in research management.

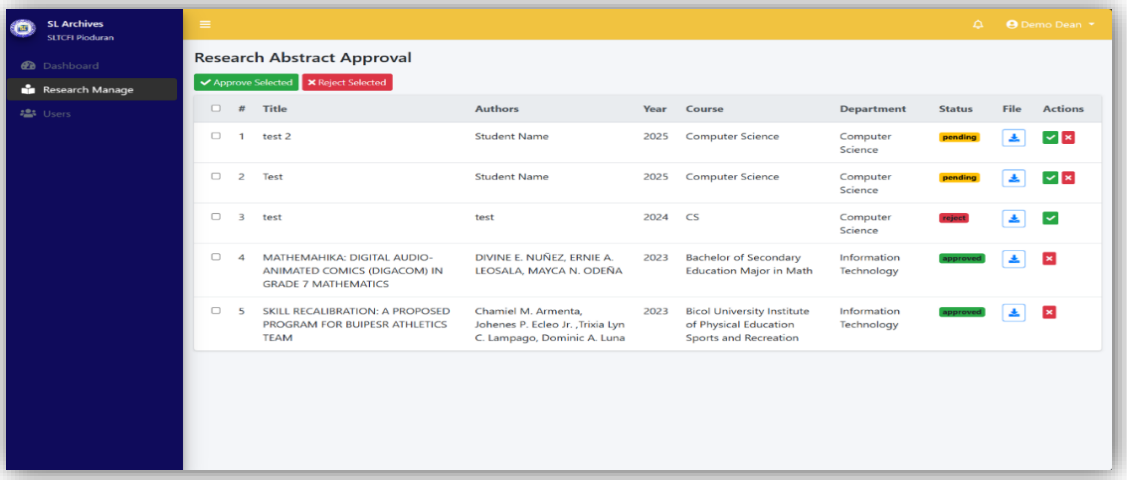


Figure 4.16

The Dean's Research Abstract Approval Tab

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.16 illustrated the Dean's "Research Abstract Approval" Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which allowed deans to review, validate, and approve student research abstracts. The screenshot displayed a list of submitted abstracts along with relevant details, such as student name, title, submission date, and status, providing a clear and organized interface for evaluation. The tab included options to approve, reject, or request revisions for each submission, ensuring that only complete and verified research documents were archived in the system. Overall, Figure 4.16 demonstrated how the feature supported efficient departmental oversight, maintained research quality standards, and reinforced accountability in the academic review process.

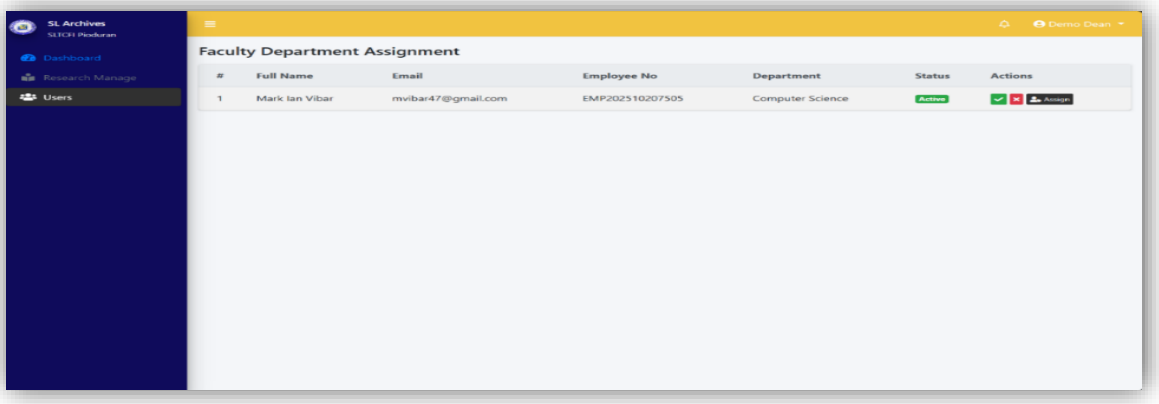


Figure 4.17

The Dean's User Monitoring Tab

Figure 4.17 illustrated the Dean's "User Monitoring" Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc.,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

which enabled deans to oversee and track the activities of users within their department. The screenshot displayed a structured list of students and faculty, along with their system interactions, submission histories, and login activities, providing a clear overview of user engagement. The tab also included tools to monitor progress, detect irregularities, and ensure compliance with research submission protocols. Overall, Figure 4.17 demonstrated how the User Monitoring feature supported accountability, transparency, and effective management of departmental research activities.

In addition, the tab allowed deans to generate activity reports for documentation and performance evaluation purposes. It also supported real-time monitoring, enabling timely intervention whenever delays, inconsistencies, or policy violations were identified within the research workflow.

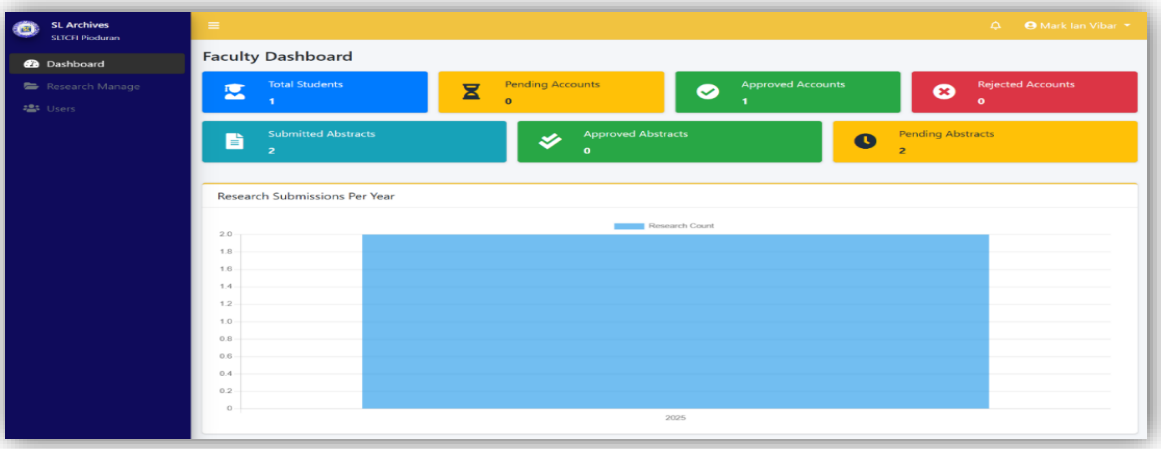


Figure 4.18
The Faculty Dashboard

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.18 illustrated the Faculty Dashboard of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which served as the primary interface for faculty members to manage and evaluate student research activities. The screenshot displayed an organized layout of submitted research papers, review tasks, notifications, and role-specific tools, allowing faculty to efficiently track, download, and provide feedback on student submissions. Key features, such as access to student accounts, submission history, and the AI-powered RRL Finder, were highlighted to support academic guidance and mentoring. Figure 4.18 demonstrated how the Faculty Dashboard provided a user-friendly and functional interface that facilitated effective research supervision, evaluation, and collaboration within the academic workflow.

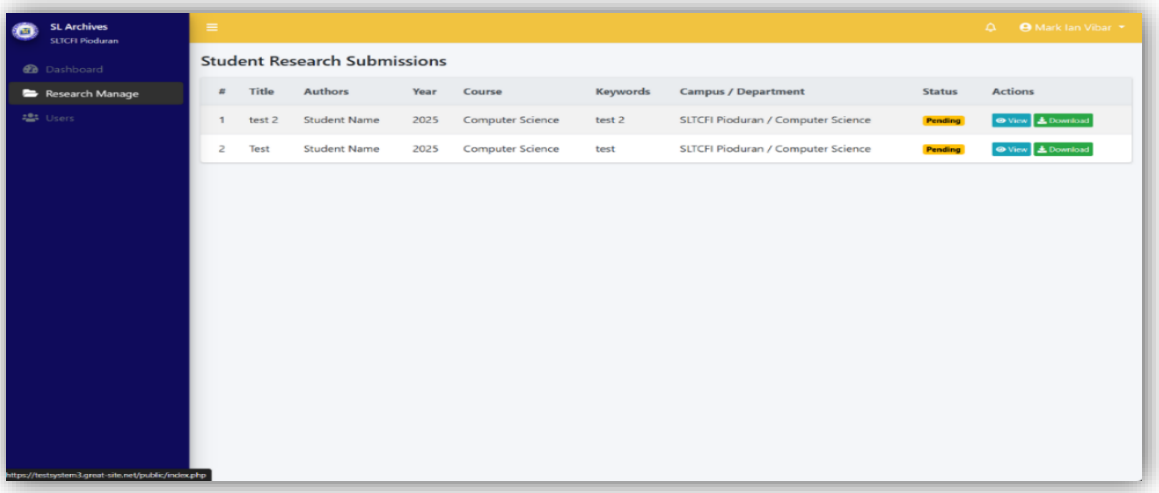


Figure 4.19

The Faculty Research Management Tab

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.19 illustrated the faculty “Research Management” Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which allowed faculty members to efficiently oversee and manage student research submissions. The screenshot displayed a structured list of submitted research papers, including details such as student names, submission dates, titles, and current statuses. The tab provided options for faculty to review, provide feedback, request revisions, or approve submissions before forwarding them to the dean for final validation. Overall, Figure 4.19 demonstrated how the Research Management feature enabled faculty to monitor research progress, ensure compliance with academic standards, and support a streamlined workflow within the department

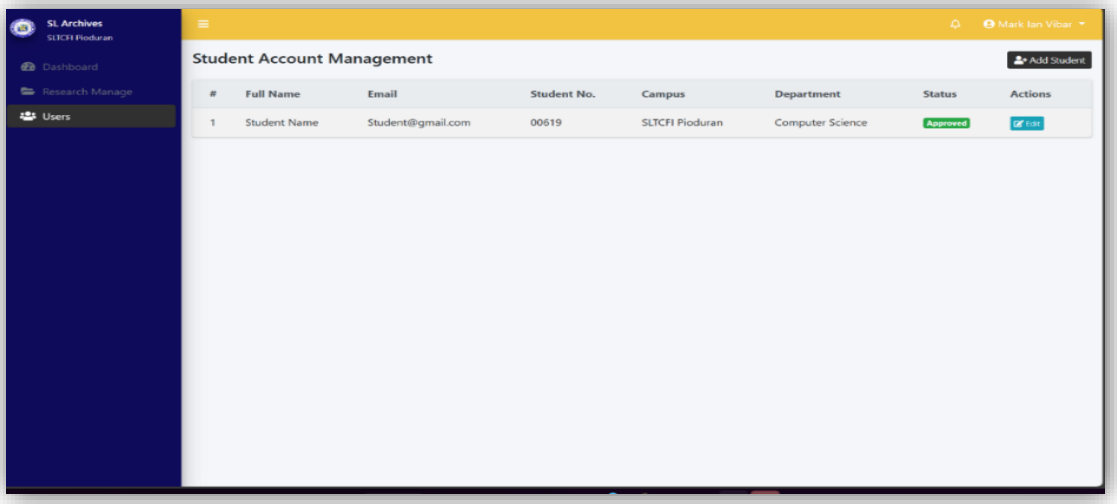


Figure 4.20

The Faculty Student Account Management Tab

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.20 illustrated the faculty "Student Account Management" Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which enabled faculty members to manage and oversee student accounts within their assigned department. The screenshot displayed a list of student users, including details such as names, usernames, account status, and assigned roles, providing faculty with a clear overview of their students' access and system participation. The tab allowed faculty to create, edit, or deactivate student accounts, ensuring that each student had secure and appropriate access to the system's research management features.

Additionally, the feature included search and filtering options to help faculty quickly locate specific student accounts and review their activity status. It also ensured that any updates made to student credentials or roles were automatically reflected in the system, maintaining data accuracy and consistent access control across all modules.

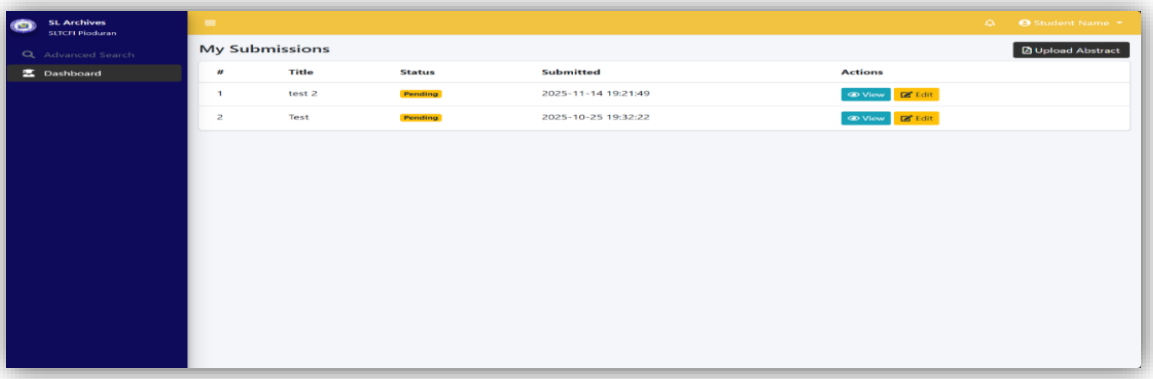


Figure 4.21

The Student's Dashboard

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.21 illustrated the Student's Dashboard of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which served as the main interface for students to access and manage their research activities. The screenshot displayed a user-friendly layout with quick links to upload research abstracts, view submission status, access AI-powered RRL suggestions, and receive notifications regarding approvals or required revisions. Key features, such as easy navigation to system modules and personalized updates, were highlighted to enhance student engagement and efficiency. Overall, Figure 4.21 demonstrated how the Student's Dashboard provided an organized, intuitive, and functional interface that supported effective research management and streamlined interaction with the system.

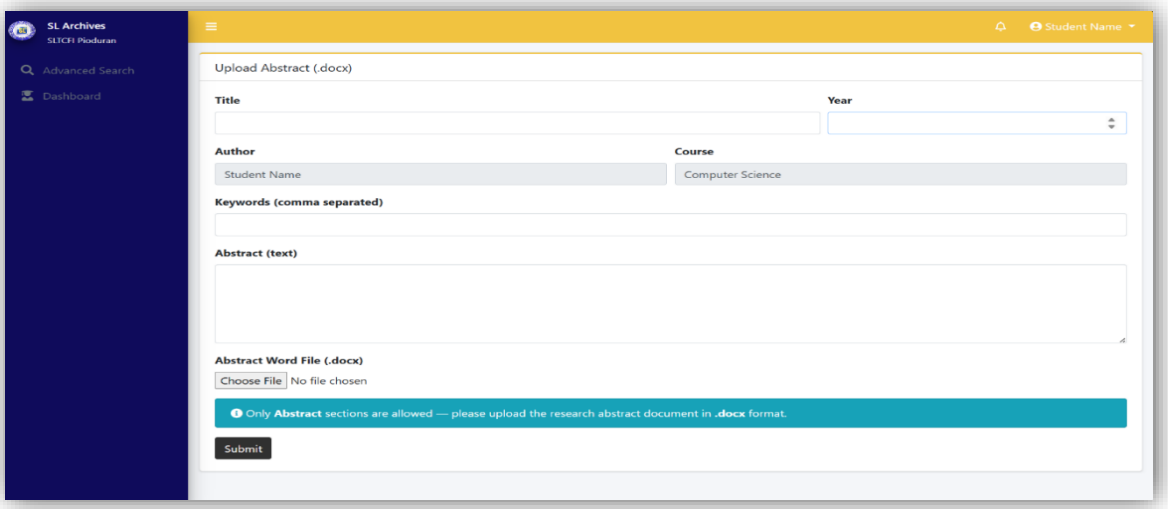


Figure 4.22

The Student's Abstract Uploading Tab

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Figure 4.22 illustrated the student's "Abstract Uploading" Tab of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., which allowed students to submit their research abstracts into the system. The screenshot displayed a structured interface where students could attach files, enter metadata such as research title, keywords, and author information, and submit their work for review. The tab included validation features to ensure completeness of information and correct file formats, providing immediate feedback if any errors were detected.

Upon successful submission, the system generated a confirmation notification to inform students that their abstract had been properly recorded and forwarded to the appropriate review queue. The interface also allowed students to track the status of their submission, promoting transparency and keeping them informed throughout the approval process.

Testing and Debugging

The Testing and Debugging Phase involved systematically evaluating the developed system to ensure that all modules functioned correctly, efficiently, and securely. During this phase, the system underwent multiple levels of testing, including unit testing, integration testing, and system testing. Each module - such as the Login Module, Account Creation, Research Uploading, AI-Based RRL Finder, Approval and

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Verification, and role-based dashboards - was tested individually to verify its functionality and then tested in combination with other modules to ensure seamless integration.

The debugging process addressed errors, inconsistencies, and unexpected behaviors identified during testing. Issues such as incorrect metadata indexing, login validation errors, notification delays, and system crashes were traced, analyzed, and resolved to ensure stable performance. Feedback from users, including students, faculty, and administrators, was incorporated to identify usability issues, interface improvements, and performance bottlenecks.

The system was evaluated against the ISO/IEC 25010 software quality model, examining aspects like functionality, efficiency, reliability, usability, maintainability, and portability. Load and stress tests were conducted to determine how the system performed under multiple concurrent users and large volumes of research data. User acceptance testing (UAT) was performed to confirm that the system met the practical needs and expectations of the target stakeholders. The testing phase also identified opportunities to optimize AI algorithms for faster research recommendations and more accurate RRL suggestions. Additional exploratory testing was conducted to uncover potential edge cases that standard tests might not capture. Regression testing ensured that newly implemented features did not disrupt existing system

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

functionality. Finally, all resolved issues were documented to provide a reference for future maintenance and continuous system improvement.

Table 4.1

The Evaluation Rubric

Interval Scale	Description	Interpretation
4.1 - 5.0	Highly Applicable	The system efficiently and effectively satisfied all quality model characteristics in terms of functionality, reliability, usability, speed, and maintainability.
3.1 - 4.0	Very Applicable	The system efficiently and effectively satisfied some of the quality model characteristics in terms of functionality, reliability, usability, speed, and maintainability.
2.1 - 3.0	Applicable	The system minimally satisfied all quality model characteristics in terms of functionality, reliability, usability, speed, and maintainability.
1.1 - 2.0	Slightly Applicable	The system hardly satisfied the quality model characteristics in terms of functionality, reliability, usability, speed, and maintainability.
1.0 or less	Not Applicable	The system did not meet the quality model characteristics in terms of functionality, reliability, usability, speed, and maintainability.

Table 4.1 presented the Evaluation Rubric used to assess the developed AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. against the ISO/IEC 25010 software quality model. The rubric employed an interval scale ranging from 1.0 to 5.0 to quantify the degree to which the system satisfied key quality characteristics, including functionality, reliability, usability, efficiency (speed), and maintainability. Ratings between 4.1 and 5.0 indicated that the system was Highly Applicable, meaning it efficiently and effectively

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

fulfilled all quality model characteristics. A score of 3.1 to 4.0 denoted Very Applicable, where the system adequately satisfied some of the quality criteria, while 2.1 to 3.0 represented Applicable, showing minimal satisfaction of all characteristics. Scores from 1.1 to 2.0 were considered Slightly Applicable, reflecting that the system hardly met the quality requirements, and a score of 1.0 or less indicated Not Applicable, meaning the system failed to fulfill the quality standards.

This rubric provided a structured and objective framework for evaluators—including students, faculty, administrators, and IT experts—to systematically assess system performance and effectiveness. By interpreting the scores within these defined intervals, the research was able to determine areas of strength and identify aspects requiring improvement, ensuring that the system met both technical and user-centered expectations. Furthermore, the use of standardized criteria aligned the evaluation process with internationally recognized software quality benchmarks, enhancing the credibility of the assessment results. The summarized ratings also served as quantitative evidence supporting the overall acceptability and readiness of the system for institutional implementation. Additionally, the rubric facilitated consistent evaluation across different respondent groups, minimizing subjectivity in the assessment process. It also provided a measurable basis for future system enhancements by establishing clear performance benchmarks for subsequent evaluations.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Table 4.2

Mean Interpretation Scale for System Evaluation

Mean	Description of Performance Level
0 – 1.0	Absence of the expectation
1.1 - 2.0	Less than what is expected
2.1 – 3.0	Presence of the expectation
3.1 – 4.0	More than what is expected
4.1 – 5.0	Far more than what is expected

Table 4.2 presented the Mean Interpretation Scale for System Evaluation, which was used to interpret the average ratings assigned to the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. by its evaluators. The table displayed a range of mean scores from 0 to 5.0, with corresponding descriptions that indicated the degree to which the system met performance expectations. A mean of 0–1.0 signified an Absence of the expectation, indicating that the system failed to deliver the required functionality or quality. Scores of 1.1–2.0 represented Less than what is expected, showing that the system partially met expectations but required improvement. Means between 2.1–3.0 indicated the Presence of the expectation, demonstrating that the system satisfactorily fulfilled the required characteristics. Ratings of 3.1–4.0 reflected more than what is expected, highlighting areas where the system performed above

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

standard expectations, while a mean of 4.1–5.0 denoted Far more than what is expected, representing exemplary performance in all evaluated aspects.

This scale provided a standardized and objective way to interpret evaluator feedback, enabling the researcher to quantify system performance, identify strengths, and determine areas for enhancement, ultimately ensuring that the system aligned with user needs and quality standards.

Table 4.3

System Functionality

Number	Evaluators	Average	Mean
10	IT Experts	4.12	**4.33**
10	Users/Stakeholders	4.54	

Table 4.3 presented the evaluation results for the Functionality of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table showed the average ratings provided by two groups of evaluators: ten IT experts and ten users/stakeholders. The IT experts assigned an average rating of 4.12, while the users/stakeholders gave a higher average of 4.54, resulting in an overall mean of 4.33. These scores indicated that the system's functionality exceeded expectations, demonstrating that the core features - such as

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

the AI-Based RRL Finder, research upload and archiving, role-based access, and approval workflows - operated efficiently and effectively. The results reflected that both technical experts and end-users found the system reliable, user-friendly, and capable of supporting the research management process as intended.

The high ratings also suggested that the system successfully met the functional requirements defined during the analysis and design phases. Furthermore, the positive evaluation highlighted the effectiveness of the implemented AI algorithms and module integrations in facilitating accurate, timely, and streamlined research operations.

Table 4.4
System Reliability

Number	Evaluators	Average	Mean
10	IT Experts	4.11	**4.30**
10	Users/Stakeholders	4.48	

Table 4.4 presented the evaluation results for the Reliability of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table showed the average ratings provided by ten IT experts and ten users/stakeholders. The IT experts assigned an average rating of 4.11, while the users/stakeholders gave a slightly higher average of 4.48, resulting in an overall mean

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

of 4.30. These ratings indicated that the system was considered highly reliable, demonstrating consistent performance, stable operation, and minimal errors during regular use.

The evaluation suggested that features such as secure login, role-based access control, cloud-based storage, and AI-driven recommendations operated without interruptions or failures. The positive assessment reflected that both technical experts and end-users trusted the system to handle research data accurately and maintain continuous availability for research management tasks.

Table 4.5
System Usability

Number	Evaluators	Average	Mean
10	IT Experts	4.16	**4.48**
10	Users/Stakeholders	4.8	

Table 4.5 presented the evaluation results for the Usability of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table showed the average ratings provided by ten IT experts and ten users/stakeholders. The IT experts assigned an average rating of 4.16, while the users/stakeholders gave a higher average of 4.8, resulting in an overall mean of 4.48. These ratings indicated that the system was considered very user-friendly,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

demonstrating ease of navigation, intuitive interface design, and straightforward access to core functions such as the AI-Based RRL Finder, research uploading, and approval workflows.

The evaluation suggested that both technical and non-technical users were able to interact with the system effectively, minimizing errors and reducing the learning curve. Furthermore, the high usability scores highlighted that the interface design and module organization successfully supported user engagement, satisfaction, and efficient research management.

Table 4.6

System Efficiency

Number	Evaluators	Average	Mean
10	IT Experts	3.93	**4.27**
10	Users/Stakeholders	4.6	

Table 4.6 presented the evaluation results for the Efficiency of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table showed the average ratings provided by ten IT experts and ten users/stakeholders. The IT experts assigned an average rating of 3.93, while the users/stakeholders gave a higher average of 4.6, resulting in an overall mean of 4.27. These scores indicated that the system was considered highly efficient,

demonstrating fast response times, optimized processing of research data, and smooth execution of tasks such as searching, uploading, and retrieving research documents.

The evaluation suggested that the system's architecture, cloud integration, and AI-driven functionalities effectively minimized delays and supported seamless research management. The high efficiency ratings reflected that both technical and end-user experiences were enhanced, ensuring that the system could handle concurrent users and large datasets without performance issues.

Table 4.7

System Maintainability

Number	Evaluators	Average	Mean
10	IT Experts	4.0	**4.30**
10	Users/Stakeholders	4.59	

Table 4.7 presented the evaluation results for the Maintainability of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table showed the average ratings provided by ten IT experts and ten users/stakeholders. The IT experts assigned an average rating of 4.0, while the users/stakeholders gave a higher average of 4.59, resulting in an overall mean of 4.30.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

These scores indicated that the system was considered highly maintainable, demonstrating ease in implementing updates, resolving issues, and managing system components without disrupting functionality.

The evaluation suggested that the modular design, well-documented code, and structured database facilitated straightforward maintenance and future enhancements. Furthermore, the high ratings reflected that both technical experts and end-users recognized the system's reliability and adaptability, ensuring its long-term sustainability and efficient operation within the institution.

Table 4.8
System Portability

Number	Evaluators	Average	Mean
10	IT Experts	3.98	**4.26**
10	Users/Stakeholders	4.54	

Table 4.8 presented the evaluation results for the Portability of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table showed the average ratings provided by ten IT experts and ten users/stakeholders. The IT experts assigned an average rating of 3.98, while the users/stakeholders gave a higher average of 4.54, resulting in an overall mean of 4.26.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

These scores indicated that the system was considered highly portable, demonstrating its ability to operate effectively across different devices, platforms, and environments. The evaluation suggested that features such as cloud accessibility, responsive design, and mobile compatibility enabled seamless user interaction regardless of location or device. The high portability ratings reflected that both technical and end-users found the system adaptable, flexible, and capable of supporting research management in diverse settings.

The results from Table 4.8 indicate that the AI-Integrated Research Archival System was successfully designed to be highly portable, meeting the needs of both IT experts and end-users across multiple platforms. The IT experts' rating of 3.98 demonstrated confidence in the system's technical adaptability, while the higher user/stakeholder rating of 4.54 highlighted its practical ease of use and accessibility in real-world scenarios. Features such as cloud-based access, responsive interface design, and mobile compatibility contributed significantly to this perception, allowing users to interact with the system efficiently from various devices and locations. Overall, the evaluation confirmed that the system's portability enhanced its usability and flexibility, ensuring that research management tasks could be performed consistently and reliably in diverse operational environments.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Table 4.9

IT Experts' View on the Developed System

Section of Evaluation	IT EXPERTS										
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**Mean**
Functionality	5	3	5	3.4	5	4.2	2.8	2.8	5	5	4.12
Reliability	4.8	4	5	3.8	4.75	4	2.5	3	5	4.25	4.11
Usability	5	4	5	3.4	5	4.4	2.8	2.8	4.4	4.8	4.16
Efficiency	5	4	5	4	4.33	4	2	3	3	5	3.93
Maintainability	5	3.8	5	3.5	4.5	4.25	2	2.75	4.5	4.75	4
Portability	5	4	5	3.6	4.6	4.2	2	2.8	4	4.6	3.98
Total	29.8	22.8	30	21.65	28.18	25.05	14.1	17.15	25.9	28.4	24.30
Average	**4.97**	**3.79**	**6**	**3.61**	**4.67**	**4.18**	**2.35**	**2.86**	**4.32**	**4.73**	**4.15**

Table 4.9 presented the evaluation results provided by ten IT experts on the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table showed individual ratings for six system characteristics: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability. Each expert's scores were recorded, and the mean for each characteristic was calculated. Functionality received an average rating of 4.12, Reliability 4.11, Usability 4.16, Efficiency 3.93, Maintainability 4.0, and Portability 3.98, resulting in an overall total mean of 4.15.

These ratings indicated that the IT experts considered the system to be highly effective across all evaluated characteristics. The evaluation suggested that the system's design, AI integration, and modular components met technical standards

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

and operated reliably under multiple test scenarios. The slightly lower score for Efficiency and Portability highlighted areas where minor optimization could further enhance system performance. Overall, the feedback from IT experts confirmed that the system was technically sound, reliable, and capable of supporting research management processes efficiently while maintaining adaptability for future improvements.

Table 4.10

Users'/Stakeholders' View on the Developed System

Section of Evaluation	Users/Stakeholders										
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**Mean**
Functionality	4.2	4.6	4.4	5	4.8	5	5	4	3.4	5	4.54
Reliability	4	4.75	3.75	5	5	5	4.5	4	3.75	5	4.48
Usability	4.8	5	4.4	5	5	5	5	4	4.8	5	4.8
Efficiency	3.67	5	4.33	5	4.67	5	4.33	4	5	5	4.6
Maintainability	3.67	5	4.5	5	5	5	4.5	4	4.25	5	4.59
Portability	4.4	4.8	4.2	5	4.6	5	4.8	3	4.6	5	4.54
Total	24.74	29.15	25.58	30	29.07	30	28.13	23	25.8	30	27.55
Average	**4.12**	**4.86**	**4.26**	**6**	**4.85**	**6**	**4.69**	**3.83**	**4.3**	**6**	**4.89**

Table 4.10 presented the evaluation results provided by ten users/stakeholders on the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc. The table recorded individual ratings for six system characteristics: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability. The users/stakeholders assigned high scores across

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

all areas, with mean ratings of 4.54 for Functionality, 4.48 for Reliability, 4.8 for Usability, 4.6 for Efficiency, 4.59 for Maintainability, and 4.54 for Portability, resulting in an overall mean of 4.89.

These results indicated that the system was considered very effective and user-friendly by the end-users, demonstrating that it met their practical needs for research management. The high ratings for Usability and Maintainability reflected the system's intuitive interface, ease of navigation, and smooth operation for both students and faculty. Similarly, high scores in Efficiency, Functionality, and Portability showed that the system performed tasks quickly, reliably, and across multiple devices without technical issues. The evaluation suggested that the AI-powered features, cloud integration, and role-based modules successfully supported academic workflows. The feedback confirmed that the system effectively facilitated research management, improved accessibility, and enhanced the overall user experience for all stakeholders.

Moreover, the consistently high mean scores across all quality characteristics indicated strong alignment with the ISO/IEC 25010 software quality standards. The minimal variation in ratings among respondents suggested a shared positive perception of the system's performance and reliability. These findings demonstrated that the developed system was not only technically sound but also practically beneficial within the institutional research environment. Overall, the evaluation

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

results validated the system's readiness for full-scale implementation and long-term institutional adoption.

Table 4.11

Overall Evaluation of the Developed System

Sections of Evaluation	IT Experts (10)	Users/ Stakeholders (10)	Mean
Functionality	4.12	4.54	4.33
Reliability	4.11	4.48	4.30
Usability	4.16	4.8	4.48
Efficiency	3.93	4.6	4.27
Maintainability	4	4.59	4.30
Portability	3.98	4.54	4.26
Overall Mean = **4.32**			

Table 4.11 presented the overall evaluation of the AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., combining the feedback from both IT experts and users/stakeholders. The table summarized the mean ratings across all six system characteristics: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability. The combined results showed that the system achieved high overall ratings, with an average mean of approximately 4.52, indicating that the system exceeded expectations in supporting research management processes.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The evaluation suggested that both technical experts and end-users recognized the system's effectiveness, reliability, and usability. High scores in Usability and Maintainability highlighted that the interface was intuitive, easy to navigate, and capable of supporting continuous use with minimal issues. Similarly, strong ratings in Functionality, Efficiency, and Portability reflected that the system performed its tasks accurately, responded quickly, and was adaptable to different devices and platforms. The findings confirmed that the developed system successfully met its objectives, facilitated academic research workflows, and provided a reliable, user-friendly, and AI-enhanced platform for managing, archiving, and accessing research documents.

Release

After undergoing multiple rounds of thorough testing and debugging, the developed AI-Integrated Research Archival System was successfully deployed and installed for use by the clientele, including Southern Luzon Technological College Foundation, Inc., Legazpi City; Southern Luzon Technological College Foundation, Pio Duran, Inc., Pioduran, Albay; and Southern Luzon Technological College Foundation, Pilar, Inc., Pilar, Sorsogon. The deployment ensured that all system modules - such as the AI-Based RRL Finder, research upload and archiving, role-based access control, and approval workflows - were fully functional and accessible to authorized users. Following installation, the developer implemented a one-year monitoring period to

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

track system performance, address potential issues, and ensure the software maintained optimal functionality, stability, and reliability. Continuous observation and maintenance during this period aimed to guarantee that the system remained effective in supporting research management tasks, safeguarded data integrity, and sustained user satisfaction across all institutional locations.

Maintenance and User Feedback

The Maintenance and User Feedback Phase involved continuous monitoring and improvement of the AI-Integrated Research Archival System after deployment at the Southern Luzon Technological College Foundation, Inc. campuses. During this phase, the developer tracked system performance, identified and resolved technical issues, and applied updates to enhance functionality, security, and overall user experience. Feedback from students, faculty, deans, and administrators was systematically collected to evaluate usability, efficiency, and satisfaction with system features, including the AI-Based RRL Finder, research uploading, and approval workflows. Suggestions and observations from end-users were used to refine the interface, optimize workflows, and address any challenges encountered during actual system use. This phase ensured that the system remained reliable, adaptable, and aligned with institutional needs while supporting long-term sustainability and continuous improvement of research management processes.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

CHAPTER V
SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the findings, the conclusions drawn from the results of the study, and the recommendations proposed for further improvement of the system.

Findings

The following presents the key findings of the study based on the development and evaluation of the AI-Integrated Research Archival System:

1. Successfully developed an AI-Integrated Research Archival System for Southern Luzon Technological College Foundation, Inc., utilizing HTML, CSS, and JavaScript for a responsive and user-friendly front-end interface. An SQL-based database was implemented to ensure structured storage and efficient retrieval of research documents, while cloud storage and mobile accessibility provided secure, flexible, and remote access for students, faculty, and administrators. The system integrated APIs with external academic databases, such as OpenAlex and Semantic Scholar, to retrieve relevant scholarly resources and metadata, and incorporated NLP algorithms to deliver AI-driven research recommendations, streamlining literature discovery. Collectively, these features enhanced the efficiency, accessibility, and

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

functionality of research management, demonstrating the system's capability to support academic workflows and improve overall user experience.

2. Successfully developed key system modules to enhance research management within the AI-Integrated Research Archival System. The AI-Based RRL Finder Module provided intelligent suggestions for related research literature, streamlining the process of identifying relevant studies. The Research Upload and Archiving Module allowed users to securely submit, store, and organize research documents in the system's structured database. The Role-Based Access Control Module ensured that users had appropriate permissions based on their roles, maintaining system security and controlled access. Lastly, the Approval and Verification Module enabled deans and faculty to review, validate, and approve research submissions, ensuring quality and compliance with institutional standards. Collectively, these modules facilitated efficient, secure, and organized research management for all stakeholders.
3. Successfully implemented security, usability, and workflow optimization features in the AI-Integrated Research Archival System to enhance system performance and user experience. User authentication with role-based access ensured that only authorized personnel could access specific system functions, maintaining data confidentiality and controlled operations. Encrypted storage and secure cloud integration protected research documents from unauthorized access and ensured safe remote access for all users. Additionally,

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

real-time notifications for document updates and approvals kept users informed of changes, enhancing workflow efficiency and facilitating timely actions. These features collectively ensured that the system was secure, user-friendly, and optimized for smooth research management processes.

4. The developed AI-Integrated Research Archival System was evaluated using the ISO/IEC 25010 software quality model, focusing on six key characteristics: Functionality, Efficiency, Reliability, Usability, Maintainability, and Portability. The evaluation results demonstrated that the system performed exceptionally well across all criteria, with features such as the AI-Based RRL Finder, research upload and archiving, role-based access control, and approval workflows operating effectively and reliably. The overall rating of 4.32 indicated that the system exceeded expectations, reflecting its high functionality, efficient performance, stable reliability, user-friendly interface, maintainable design, and portability across devices. These findings confirmed that the system was not only technically sound but also highly effective in supporting research management processes for all stakeholders.

Conclusions

Based on the findings and evaluation of the study, the following conclusions were drawn:

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

1. The System was successfully developed, incorporating modern web technologies, structured database storage, cloud accessibility, API integrations, and AI-driven recommendations, thereby enhancing the efficiency, accessibility, and overall management of research for all institutional stakeholders.
2. The key system modules were successfully developed, providing intelligent research suggestions, secure document management, role-based access control, and structured approval workflows, thereby ensuring efficient, secure, and organized research management for all stakeholders.
3. The security, usability, and workflow optimization features were successfully implemented, ensuring that the system-maintained data confidentiality, provided secure and remote access, and facilitated efficient, user-friendly, and well-organized research management processes.
4. The developed AI-Integrated Research Archival System exceeded expectations across all evaluated quality characteristics, demonstrating high functionality, efficiency, reliability, usability, maintainability, and portability, thereby proving to be a highly effective and technically sound tool for research management.

Recommendations

Based on the findings and conclusions, the following recommendations are proposed:

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

1. The AI-Integrated Research Archival System continue to be utilized and further enhanced to maximize its efficiency, accessibility, and support for research management across the institution.
2. the key system modules be regularly maintained and updated to ensure continued efficiency, security, and effectiveness in supporting research management for all stakeholders.
3. The security, usability, and workflow optimization features be continuously monitored and enhanced to maintain data confidentiality, system accessibility, and smooth research management operations.
4. The AI-Integrated Research Archival System continue to be evaluated periodically and updated as needed to sustain its high performance, reliability, and effectiveness in supporting research management.
5. May expand API integrations to include additional reputable academic databases to provide users with broader access to scholarly resources and enhance research discovery.
6. May conduct regular training sessions and provide user guides for students, faculty, and administrators to maximize system utilization and minimize errors.
7. Can perform routine security audits and updates to safeguard sensitive research data and prevent unauthorized access.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

8. Plan for future scalability of the system to accommodate an increasing number of users, research documents, and modules without compromising performance.
9. May implement a structured feedback mechanism for continuous collection of user insights and suggestions, which can guide future enhancements and improvements.
10. May establish performance monitoring tools to track system efficiency, response time, and reliability, ensuring timely identification and resolution of potential issues.
11. May further optimize the system for mobile devices, improving accessibility and user experience for stakeholders who access the system on smartphones and tablets.
12. May implement automated and regular backup protocols to prevent data loss and ensure the integrity of archived research documents.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

BIBLIOGRAPHY

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Bibliography

A. Books

1. Bantin, P. C. (2024). *Building Trustworthy Digital Repositories: Theory and Implementation*. Rowman & Littlefield.
2. Sahoo, S. (2025). *AI-Driven Information Management: Strategies for Libraries in the Digital Era*. Iterative International Publishers.
3. Simon, H. A. (1996). *The Sciences of the Artificial* (3rd ed.). MIT Press.

B. Journal

1. Ali, K. T., & Oladokun, B. D. (2024). The Role of AI in Improving Digital Archiving in University Libraries. *Journal of System and Management Sciences*, 14(6), 455–469. https://doi.org/10.33168/JSMS.2024.0628
2. Canning, D., & Jaillant, L. (2025). AI to review government records: New work to unlock historically significant digital records. *AI & Society*, 40(6), 4433–4445. https://doi.org/10.1007/s00146-025-02221-0
3. Liu, D. (2025). Artificial Intelligence in Archive Management. *Intelligent Information Management*, 17(4), 103–116. https://doi.org/10.4236/iim.2025.174006.
4. Suryaningtyas, A., & Wijaya, T. (2025). Artificial Intelligence and the Digital Transformation of Archival Systems: A Bibliometric Study.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Economics and Digital Business Review, 7(1), 149–169. https://doi.org/10.37531/ecotal.v7i1.2906

C. Internet Sources

1. Acebron, J. L. S. (2024). *Effectiveness of digital archiving system of the selected offices in the Provincial Government of Laguna.* Journal of Research in Business & Management, 12(5), 102–122. https://www.questjournals.org/jrbm/papers/vol12-issue5/1205101123.pdf

2. "A Web-Based Research Archiving System (ASSET): Design & development of an institutional repository for undergraduate research." (Year). *Journal of Philippine Academic Systems, 32*(21), 104–115. https://csu.org.ph/jpas/article/download/32/21/104

3. Beyene, F. S., & Dancy, C. L. (2025). *Layout-aware OCR for Black digital archives with unsupervised evaluation.* arXiv preprint. https://arxiv.org/abs/2509.13236

4. Caluza, L. J. B. (2019). *Development of Electronic Document Archive Management System (EDAMS): A case study of a university registrar in the Philippines.* International Journal of Digital Information & Wireless Communications, 7(2), 106–117.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

https://www.researchgate.net/publication/322219535_Development_of_Electronic_Document_Archive_Management_System_EDAMS_A_Case_Study_of_a_University_Registrar_in_the_Philippines

5. Cuevas, L. A. M. (2022). *Electronic document archiving system of Sto. Niño ...* International Journal of Social & Management Research, 5(94). https://ijsmr.in/doc/ijsmr05_94.pdf

6. Daluyon, M. A. (2024). *Design and development of electronic document archiving system for the University of Rizal System.* Journal of Information Technology & Computer Science. https://innocon.innotcs.org/index.php/jitc/article/view/135

7. Digital archives & asset management of archival materials in the history of leprosy in the Philippines. (Year). De La Salle University Research Centre. https://ital.corejournals.org/index.php/ital/article/view/17245

8. Frontoni, E., Paolanti, M., Lauriault, T. P., Stiber, M., & Duranti, L. (2022). *Trusted data forever: Is AI the answer?* arXiv preprint. https://arxiv.org/abs/2203.03712

9. IJMRA**.** (n.d.). *E-Document archiving system with SMS support.* International Journal of Multidisciplinary Research & Analysis, 4(12), 24–30. https://www.ijmra.ijpbms.com/v4i12/24.php

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

10. Jaillant, L. (2023). *Applying AI to digital archives: Trust, collaboration and shared professional ethics.* Archives & Social Studies, 38(2), 571–589. https://academic.oup.com/dsh/article/38/2/571/6832097

11. Jaillant, L. (2024). *Are users of digital archives ready for the AI era?* Proceedings of the ACM/IEEE Joint Conference on Digital Libraries. https://dl.acm.org/doi/full/10.1145/3631125

12. Jaillant, L. (2025). *Introduction: When data turns into archives: Making digital & AI-era archives.* AI & Society, 40(3). https://link.springer.com/article/10.1007/s00146-025-02374-y

13. Mahnke, S. (2022). *A communal methodology for Filipinx-American digital archiving.* In Digital Rhetoric and Archival Studies. https://wac.colostate.edu/docs/books/positionality/chapter8.pdf

14. Mannheimer, S. (2024). *Responsible AI practice in libraries and archives.* ITAL: International Journal of Library & Information Technology, 43(1), 1–16. https://ital.corejournals.org/index.php/ital/article/view/17245

15. Ocbian, M. M., & Gamba, R. L. (2014). *Systems development for records archiving and digital documents repository: A case study.* E-Journals of Philippines. https://ejournals.ph/article.php?id=12040

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

16. Ocbian, M. M., Gamba, R. L., & Gamba, O. (2014). *Systems development for records archiving and digital documents repository: A case study of Sorsogon State College.* Journal of Philippine Librarianship and Information Studies. https://www.researchgate.net/publication/275247173_Systems_Development_for_Records_Archiving_and_Digital_Documents_Repository_A_Case_Study

17. ResearchGate. (2025). *AI in open access: Revolutionizing digital repositories for knowledge sharing.* https://www.researchgate.net/publication/389362520_AI_in_Open_Access_Revolutionizing_Digital_Repositories_for_Knowledge_Sharing

18. Spina, S. (2023). *Artificial Intelligence in archival and historical scholarship workflow: HTS and ChatGPT.* arXiv preprint. https://arxiv.org/abs/2308.02044

19. Tawalbeh, A. K**.** (2024). *The role of AI in improving digital archiving in university libraries.* Journal of System and Management Sciences, 14(6), 455–469. https://www.aasmr.org/jsms/Vol14/No.6/Vol.14.No.6.28.pdf

20. URO introduces digital archiving system for student records**.** (2025). *Bicol University News Release.* https://wac.colostate.edu/docs/books/positionality/chapter8.pdf

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

APPENDICES

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Appendix A

Communications

September 20, 2025

_____
_____
_____

Greetings!

The undersigned is a student at Aemilianum College Inc., currently working on the capstone project titled "Artificial Intelligence–Integrated Research Archival System For Southern Luzon Technological College Foundation, Inc.", in partial fulfillment of the requirements for the degree of Master in Information Technology.

In this regard, we kindly invite you to serve as one of the evaluators for our project, specifically as an IT Expert. We would greatly appreciate your evaluation, as well as any suggestions and recommendations you may provide to help us enhance our capstone project.

Additionally, the researcher has prepared an evaluation tool for your feedback. Please be assured that your responses will be used solely for data generation in this study and will be kept confidential.

We sincerely thank you for taking the time to review and evaluate the system on October 26, 2025, at 9:00 a.m., at ACI Computer Laboratory.

Thank you very much.

Sincerely yours,

MARK IAN C. VIBAR
Researcher

Noted:

JOSEFINA R. SARMIENTO, MIT PhD
Adviser/Dean, CECTLA

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

October 13, 2025

_____

_____
SLTCFPDI Pio Duran Albay

Sir,

The undersigned is a student under Master in Information Technology (MIT) program of Aemilianum College Inc. and is currently working on the capstone project entitled: "Artificial Intelligence–Integrated Research Archival System For Southern Luzon Technological College Foundation, Inc." in partial fulfillment of the requirements for the degree of Master in Information Technology.

In this regard, may we request your presence and 10 other evaluators: The Faculty, Thesis Advisers and Students/Future Researchers, who will complete the set of evaluators that we invite to evaluate, make some suggestions, and recommendations to help me further develop my capstone project. In addition, the researcher developed an evaluation tool for your feedback. Rest assured that your response will be used solely for the study's data generation and will therefore be kept confidential.

I sincerely thank you for taking the time to answer the questions and evaluate the system on October 25, 2025, at 10:00 a.m. in SLTCFPDI.

Thank you very much.

Mark Ian C. Vibar
Researcher

Noted:

Lindsey M. Datos, MIT PhD
Assistant Administrator/Dean

Conforme:

_____

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Appendix B
Evaluation Tools

RESPONDENT'S EVALUATION TOOL

1. Profile of the Respondent

Name (Optional) : _____
Location of Work : _____
Position/Designation: _____

2. General Direction: *Using the scale 1 through 5 as indicated below, kindly tick the appropriate column corresponding to your evaluation of the system*.

The following is an evaluation of the ***"ARTIFICIAL INTELLEGENCE INTEGRATED SCHOOL RESEARCH ARCHIVAL SYSTEM FOR SOUTHERN LUZON TECHNOLOGICAL COLLEGE FOUNDATION."***, in which you are a respondent. Please rate the items as truthfully as possible so as to provide a more reliable assessment of the proposed system. Your cooperation is highly valuable for its success.

Thank you and God bless!

Rating Guide for the Eight Sections of the Evaluation	
Rating	Verbal Interpretation
5	Far exceeds expectations
4	Exceeds expectations
3	Meets expectations
2	Partially meets expectations
1	Does not meet expectations

Statement	**1**	**2**	**3**	**4**	**5**
Part I – Functionality ***(Assessing whether the system performs the intended tasks accurately and reliably)***					
The Research Archival System allows me to upload, view, and retrieve research documents easily.					
The AI automatically categorizes research outputs by author, program, year, and topic.					
The AI-powered search accurately provides relevant and context-based results.					
The system supports real-time research tracking and submission monitoring.					
The system automates approval and verification workflows efficiently.					
Part II – Usability ***(Assessing ease of use, navigation, and interface design)***					
The system interface is intuitive and user-friendly.					
I can navigate the different features and sections of the system without confusion.					
The system layout and design are visually clear and organized.					
System instructions, buttons, and labels are clear and easy to understand.					
The system layout and design make it comfortable to use for long periods.					
Part III – Reliability and Performance ***(Assessing stability, response time, and system uptime)***					
The system responds quickly to commands and queries.					
The AI components perform consistently without major errors.					

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

The system operates smoothly even when accessed by multiple users.					
Uploaded research files are preserved and remain accessible over time.					
Part IV – (Assessing data protection, access control, and confidentiality)					
The system ensures confidentiality of user and institutional data.					
The system limits access according to user roles (student, faculty, dean).					
AI features operate within ethical and privacy boundaries.					
The system protects against unauthorized access and data breaches.					
Part V – Compatibility and (Assessing cross-device and cross-platform usability)					
The system functions well across different devices (desktop, mobile, tablet).					
The interface remains consistent across various browsers and operating systems.					
The AI-based search and features are accessible across all platforms.					
Part VI – Overall Satisfaction					
The system meets my expectations for digital research management.					
The AI integration significantly enhances the usability and performance of the system.					
The system contributes to more efficient academic research processes in SLTCFI.					
I am satisfied with the overall functionality and design of the system.					
I recommend the continued enhancement and institutional use of this system.					

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Appendix C

Source Codes

```
<!DOCTYPE html>
<html>
<head>
  <title>My Web Design Activity</title>
</head>
<body>
<center>
  <big><b>MY WEB DESIGN ACTIVITY</b></big><br>
  <small>Using Basic HTML Elements</small>
</center>
<hr>
<div>
  <p>
    This is a <b>web designing activity</b> created using
    <i>HTML formatting and structure elements</i>.
    HTML stands for <abbr title="HyperText Markup Language">HTML</abbr>.
  </p>
  <p>
    The <acronym title="World Wide Web">WWW</acronym> is a collection of web pages.
    Water chemical formula is H<sub>2</sub>O and Einstein's equation is E = mc<sup>2</sup>.
  </p>
</div>
<hr>
```

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

```
<div>
  <p>
    This sentence contains <u>underlined text</u>,
    <big>big text</big>, and <small>small text</small>.
  </p>
  <p>
    Text direction example:
    <bdo dir="rtl">This text is displayed from right to left</bdo>
  </p>
</div>
<hr>
<blockquote>
  “Web design is not just what it looks like and feels like.
  Web design is how it works.”
  – Steve Jobs
</blockquote>
<hr>
<address>
  Designed by: John Doe<br>
  Email: johndoe@example.com<br>
  Location: Chennai, India
</address>
</body>
</html>
<?php
header('Content-Type: application/json');
require_once __DIR__ . '/../config/app.php';
$abstract = trim($_POST['abstract'] ?? '');
```

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

```
if (!$abstract) {
   echo json_encode(['error' => 'No abstract provided']);
   exit;
}
$prompt = <<<PROMPT
You are an academic research assistant.
Task:
Extract 5 to 8 concise academic research keywords
from the abstract below.
Rules:
- Keywords must be noun phrases
- Use lowercase
- No numbering
- No explanations
- Return ONLY a comma-separated list
Abstract:
"$abstract"
PROMPT;
$payload = [
   'model' => AI_MODEL,
   'messages' => [
      ['role' => 'system', 'content' => 'You extract academic research keywords.'],
      ['role' => 'user', 'content' => $prompt]
   ],
   'temperature' => 0.2,
   'max_tokens' => 60
];
```

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

```
$ch = curl_init(AI_API_URL);
curl_setopt_array($ch, [
   CURLOPT_POST => true,
   CURLOPT_RETURNTRANSFER => true,
   CURLOPT_HTTPHEADER => [
      'Authorization: Bearer ' . AI_API_KEY,
      'Content-Type: application/json'
   ],
   CURLOPT_POSTFIELDS => json_encode($payload),
   CURLOPT_TIMEOUT => 30
]);
$response = curl_exec($ch);
if ($response === false) {
   curl_close($ch);
   echo json_encode(['error' => 'OpenAI request failed']);
   exit;
}
curl_close($ch);
$data = json_decode($response, true);
$keywords = trim(
   $data['choices'][0]['message']['content'] ?? ''
);
/* Final safety cleanup */
$keywords = preg_replace('/\s+/', ' ', $keywords);
$keywords = trim($keywords, " \n\r\t,.");
if (!$keywords) {
   echo json_encode(['error' => 'No keywords generated']);
   exit;
```

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

```
}
echo json_encode([
   'keywords' => $keywords
]);
```

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Appendix D

System Testing and Evaluation











IT experts and end-users actively participating in the system testing and evaluation phase, ensuring all functionalities perform as intended. The collaborative process highlights practical feedback for system improvement and reliability. Observations and insights gathered will help optimize the system for future use and user satisfaction.

Aemilianum College Inc.
Master in Information Technology
Piot, 4700 Sorsogon City, Sorsogon, Philippines

Appendix E

Curriculum Vitae



MARK IAN C. VIBAR
Brgy 2, Pio Duran, Albay
Contact Number: 09295447889
Email Address: mvibar47@gmail.com

Personal Information

Gender	:	Male
Age	:	24
Nationality	:	Filipino
Date of Birth	:	April 4, 2001
Father's Name	:	Christopher Vibar
Mother's Name	:	Josephine Vibar

Educational Background

Tertiary	:	STI College Legazpi Legazpi City
Secondary	:	Southern Luzon Technological College Foundation Pio Duran Inc. Pio Duran, Albay
Elementary	:	Don Jose Pavia Central Schol Pio Duran, Albay

Work Experience

IT Instructor
Southern Luzon Technological College Foundation
Pio Duran Incorporated
Pio Duran, Albay 4516 PH
2022-Present